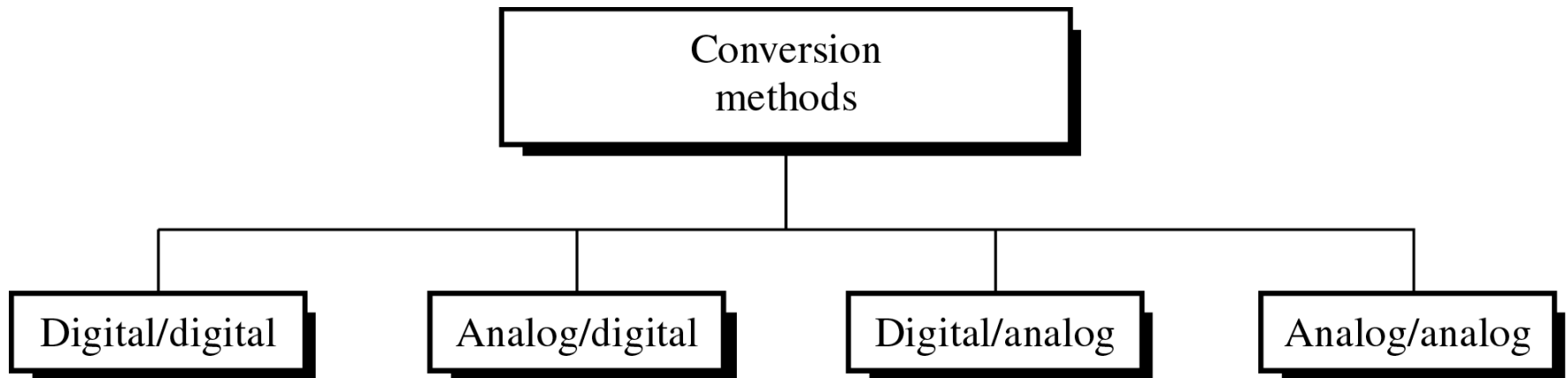


Encoding and Modulating

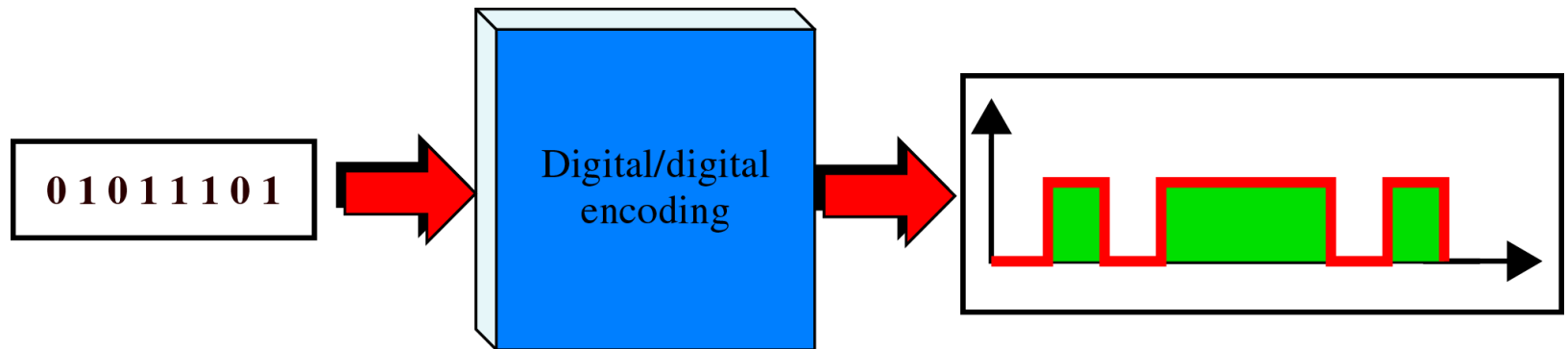
Encoding concern with the conversion of data (digital/analog) into signal (digital/analog).

We must transform data into signals to send them from one place to another.

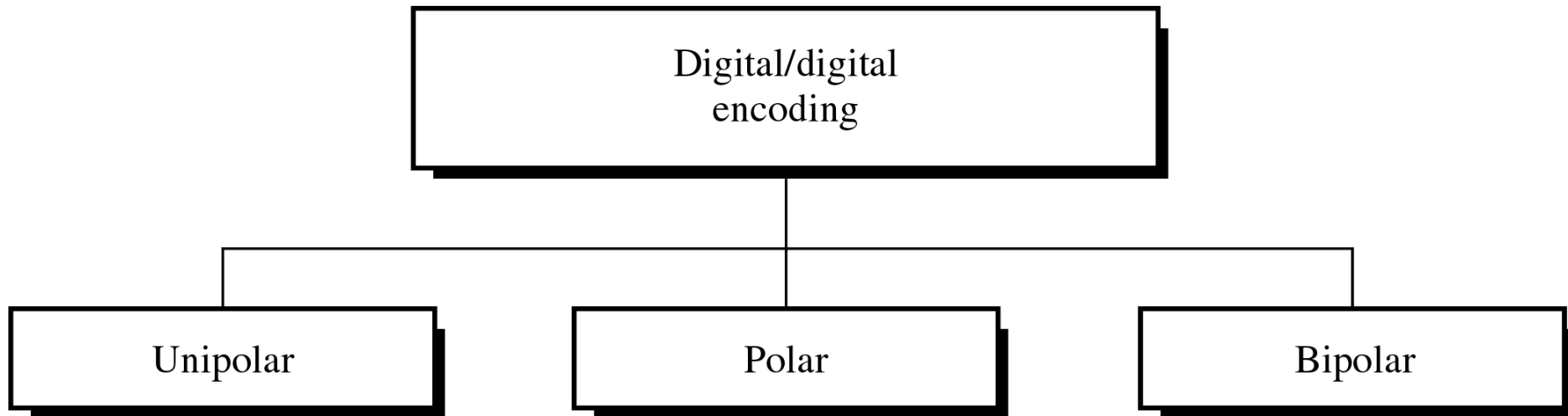
Different Conversion Schemes



Digital to Digital Encoding

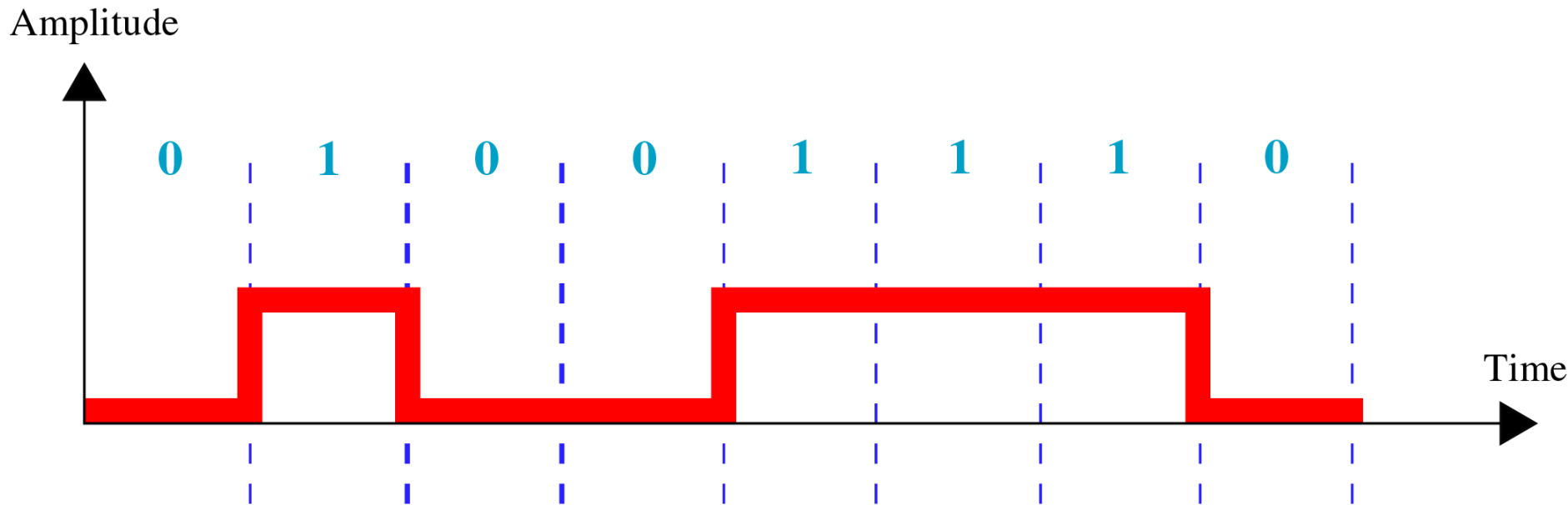


Types of Digital to Digital Encoding



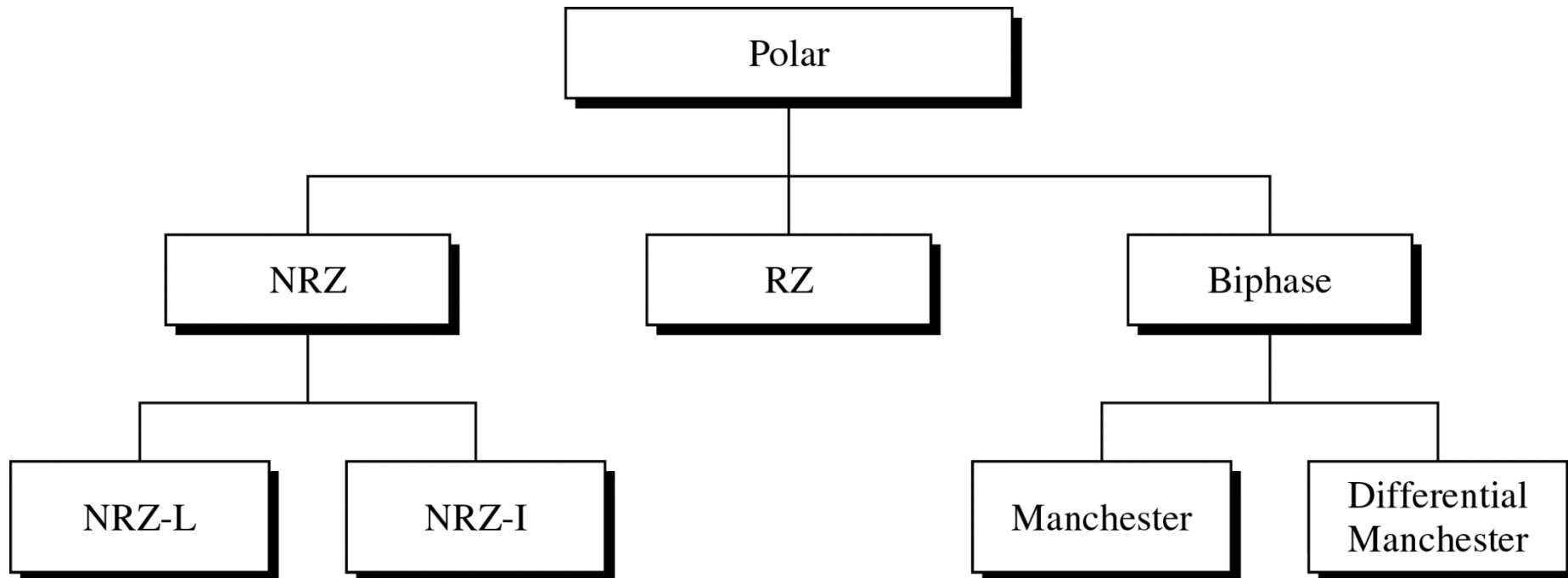
Unipolar Encoding

Uses only one level of value. 0 represent by zero voltage & 1 by positive voltage.



It has two problems: DC component and synchronization.

Types of Polar Encoding

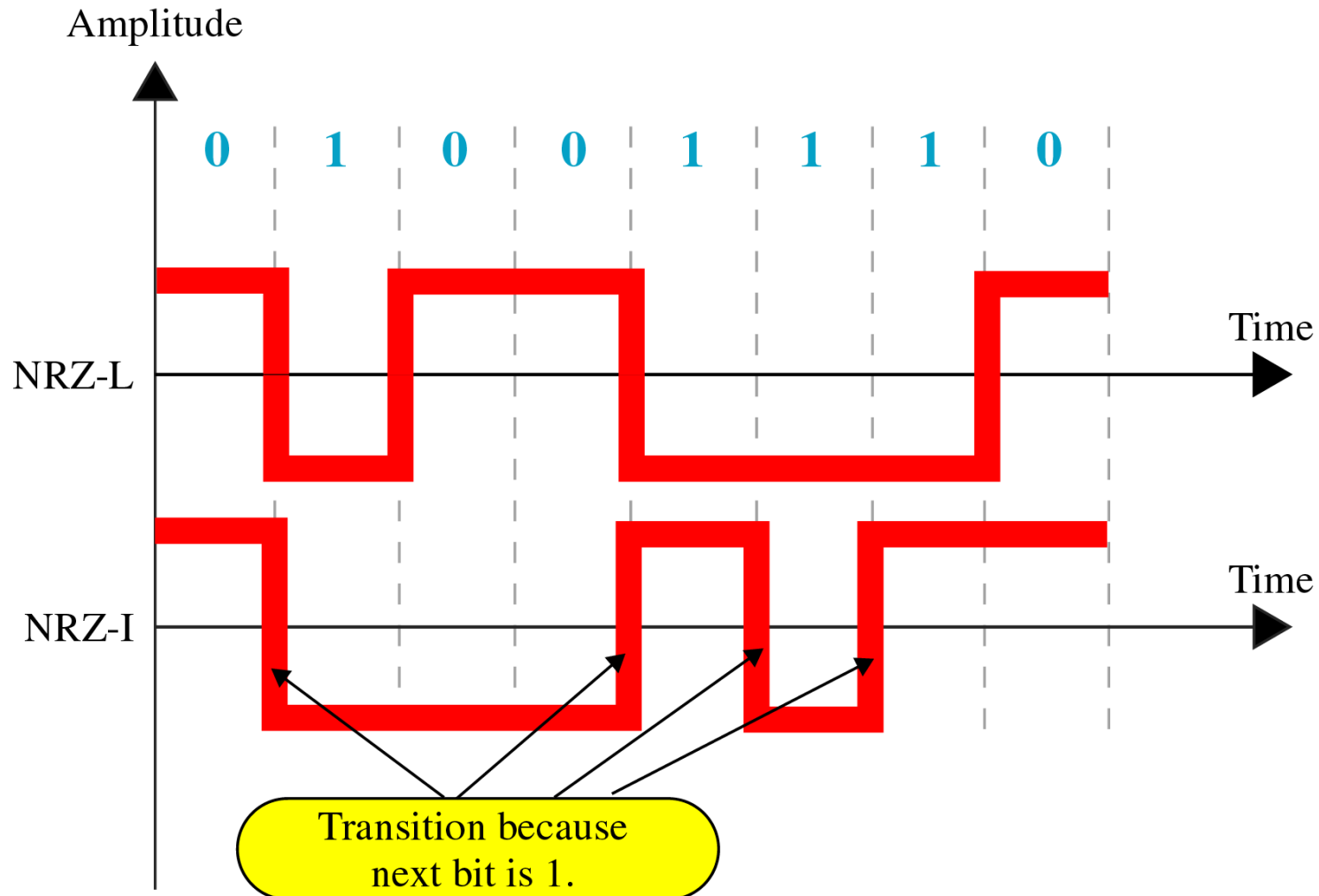


These encoding uses two levels of amplitude (+, -).

NRZ-L and NRZ-I Encoding

NRZ-L: the level of signal is depend upon the state of the bit. 0 :+ voltage and 1 : - voltage.

NRZ-I: the signal is inverted if a 1 is encountered.

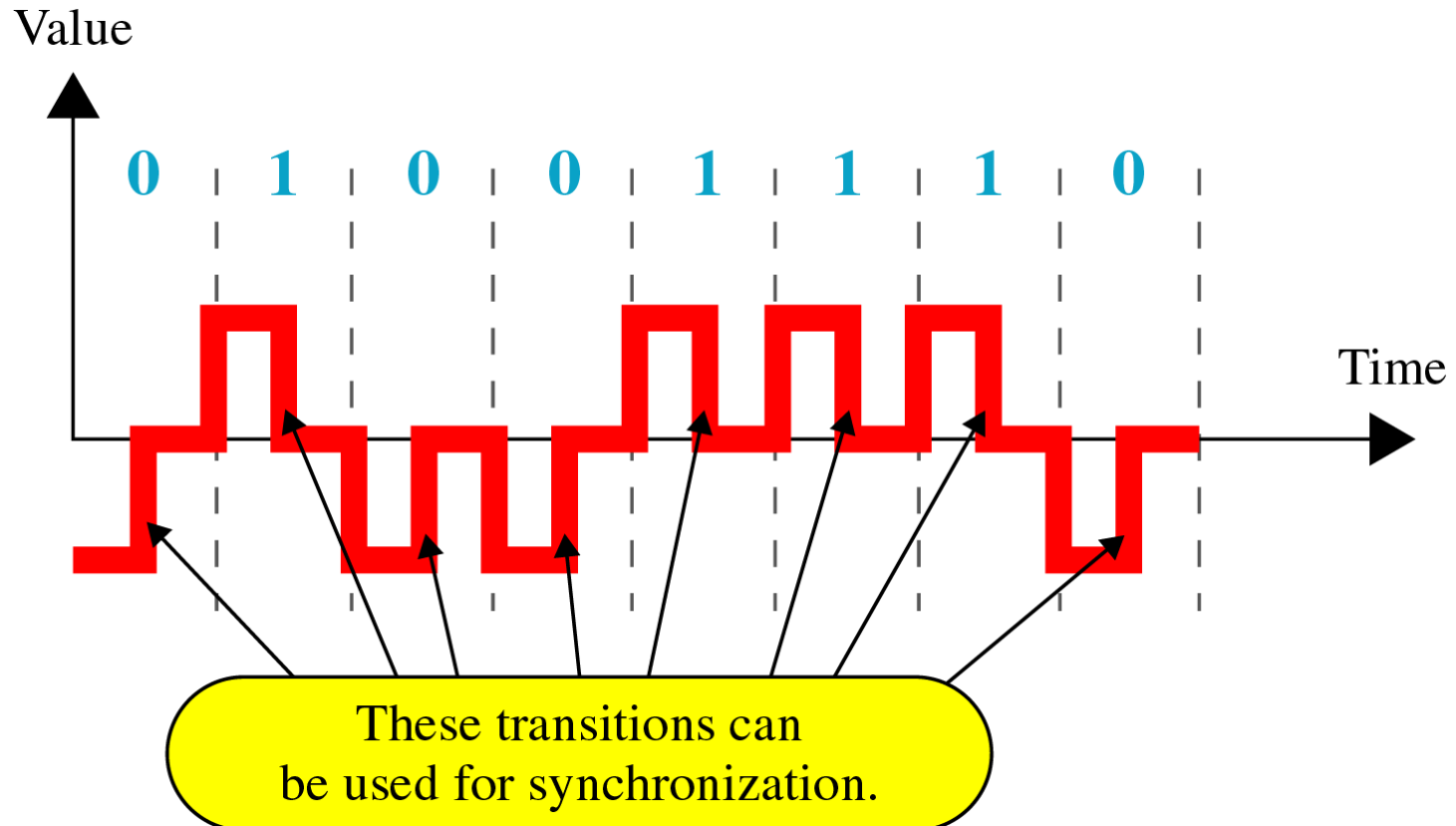


RZ Encoding

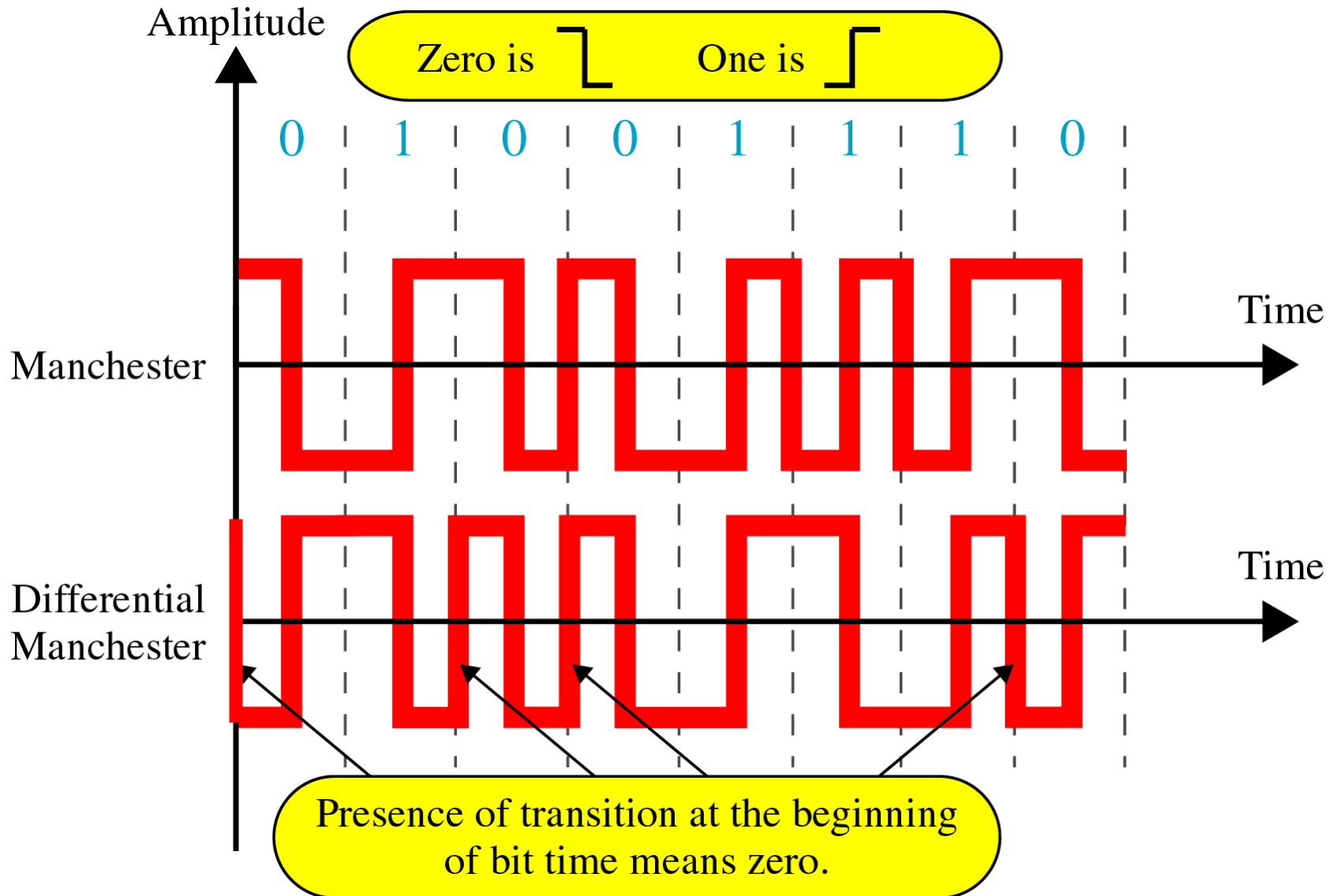
RZ uses three values +, -, and 0. The signal changes not between the bits but during the middle of each bit.

1 : positive to zero and

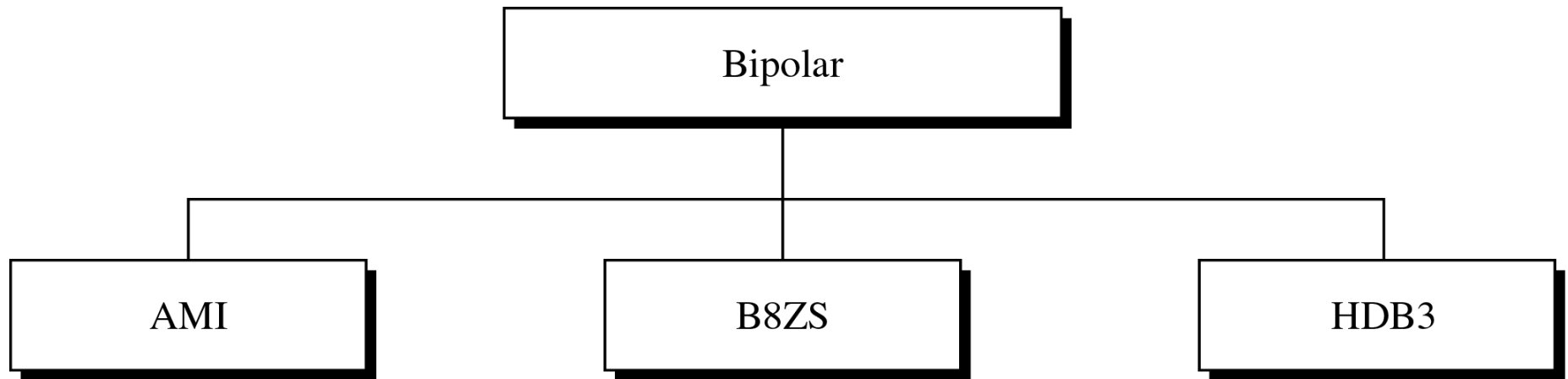
0: negative to zero



Manchester and Diff. Manchester Encoding



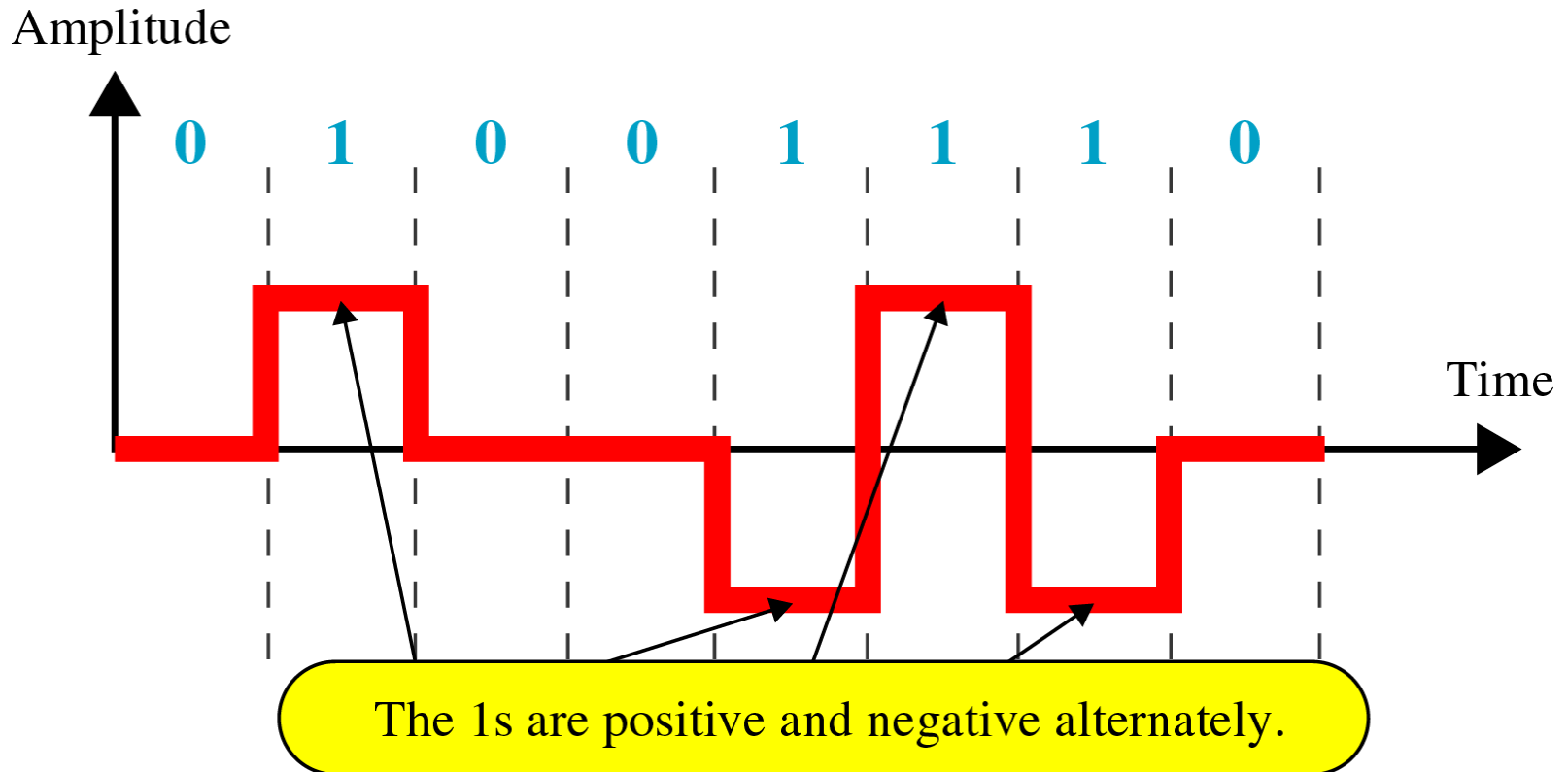
Types of Bipolar Encoding



Bipolar encoding uses three level of voltage: positive, negative and zero.

Bipolar AMI Encoding

0: zero voltage and 1 : positive and negative alternately.



Long occurrence of 0s causes synchronization problem.

Pseudoternary Encoding:

1: zero voltage and 0 : positive and negative alternately

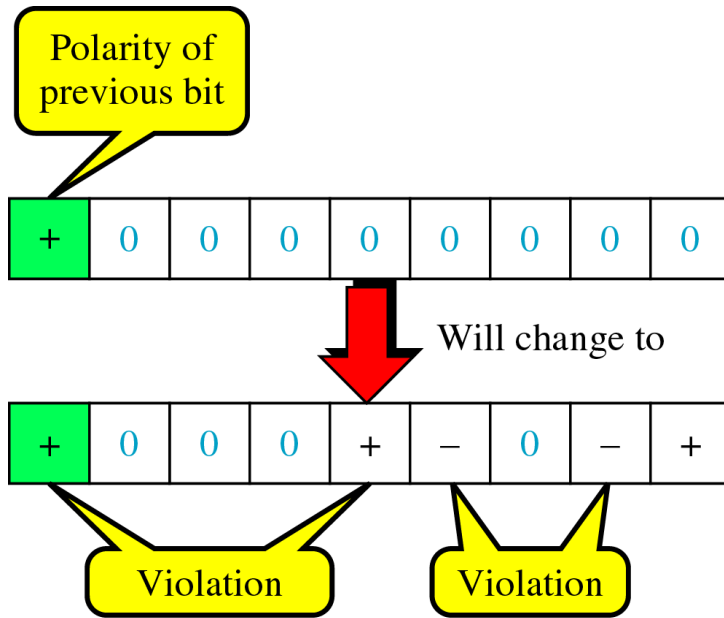
Two variations of bipolar AMI have been developed to solve the problem of synchronizing sequential 0s, especially for long-distance transmission.

- Bipolar 8 Zero Substitution (B8ZS)
- High-Density Bipolar 3 (HDB3)

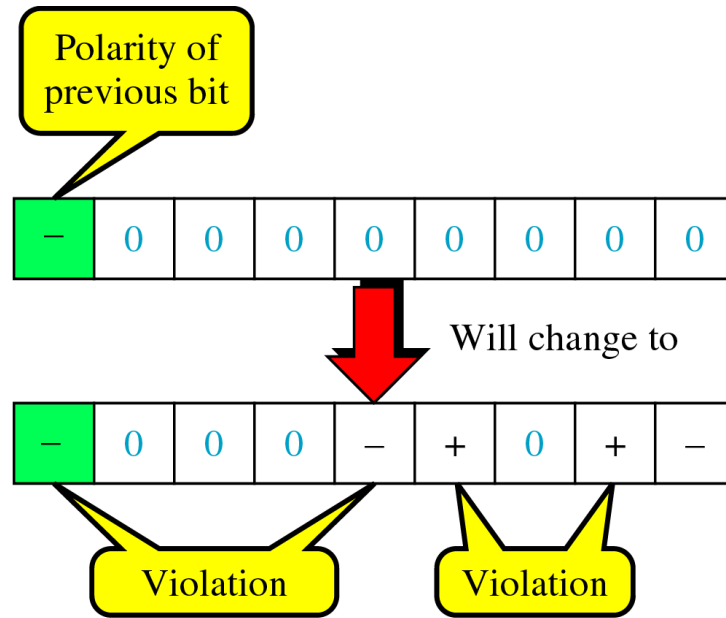
B8ZS Encoding

- In most situations, B8ZS functions identically to bipolar AMI.
- B8ZS also changes poles with every 1 it encounters.
- The difference between B8ZS and Bipolar AMI occurs whenever 8 or more consecutive 0s are encountered in the data stream.
- Whenever 8 or more consecutive 0s are encountered, B8ZS is to force artificial signal changes, called violations, within the 0 string.
- Anytime 8 zeros occur in succession, B8ZS introduces changes in the pattern one of two ways based on the polarity of previous 1 bit.

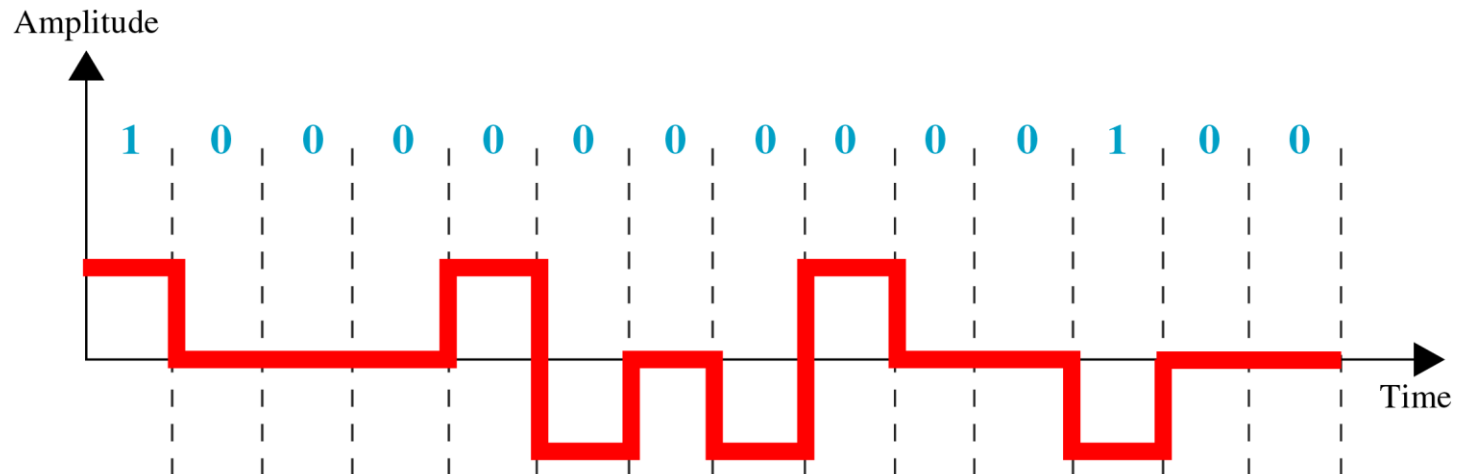
B8ZS Encoding



(a)

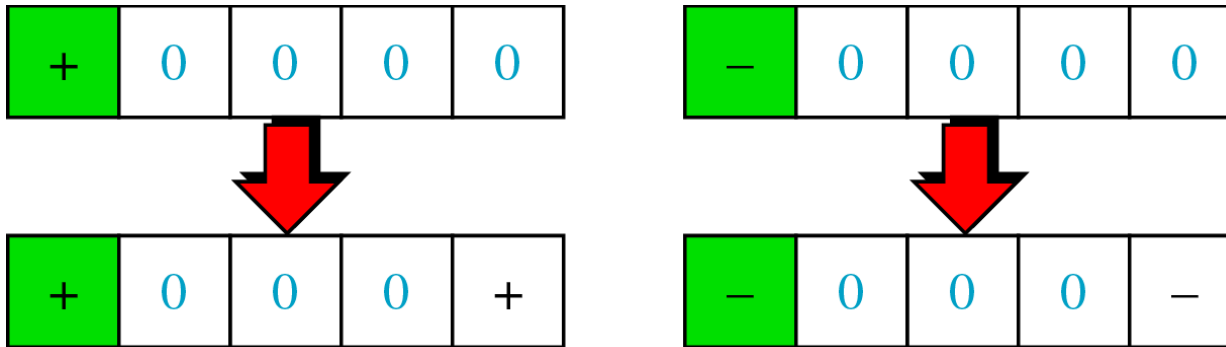


(b)

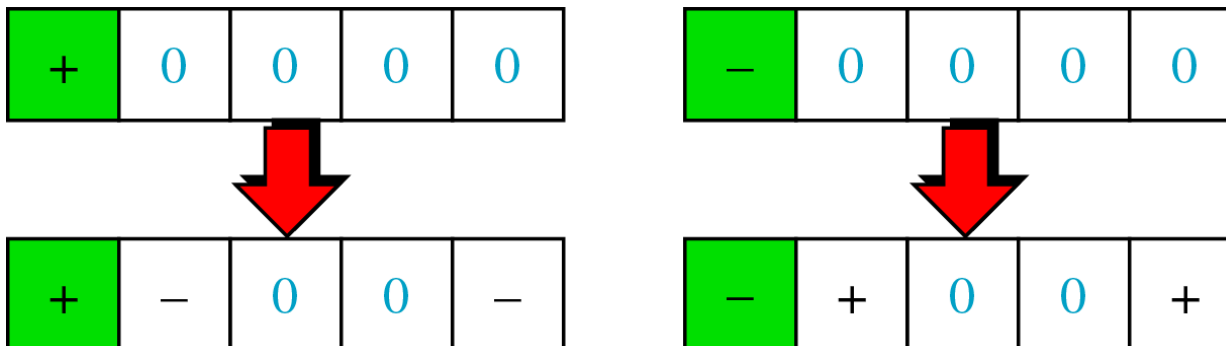


HDB3 Encoding

If four 0s come one after another, we change the pattern in one of four ways based on the polarity of the previous bit and the number of 1s since the last substitution.

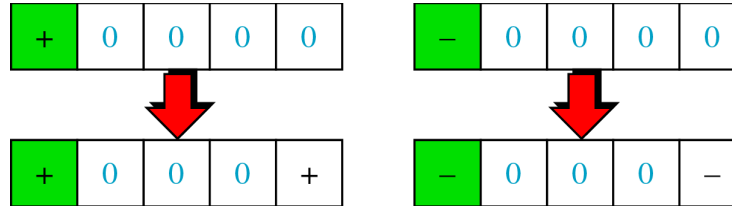


(a) If the number of 1s since the last substitution is odd

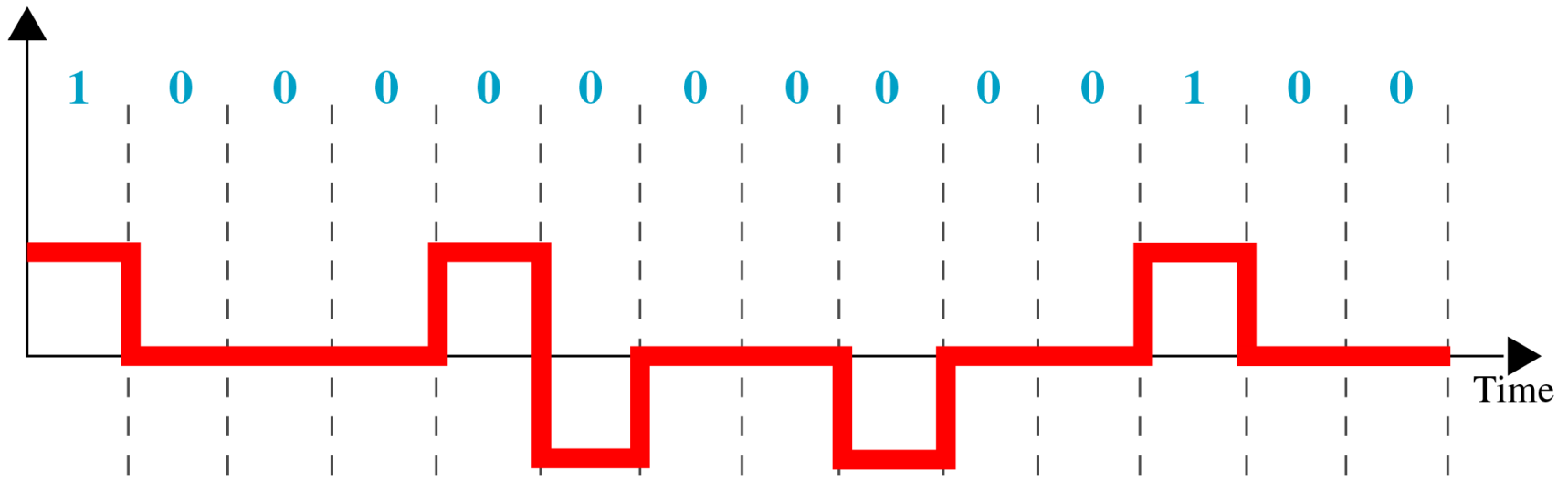


(b) If the number of 1s since the last substitution is even

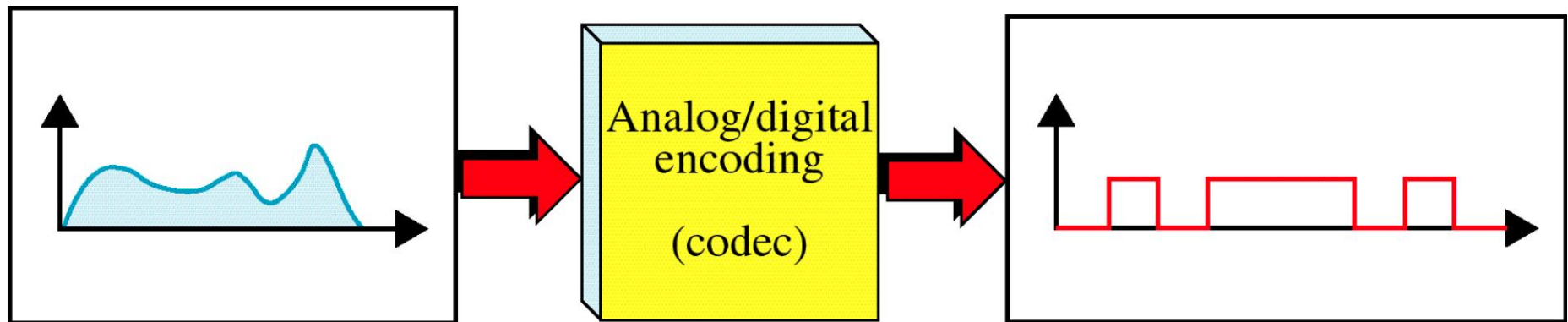
Encoding using HDB3



Amplitude

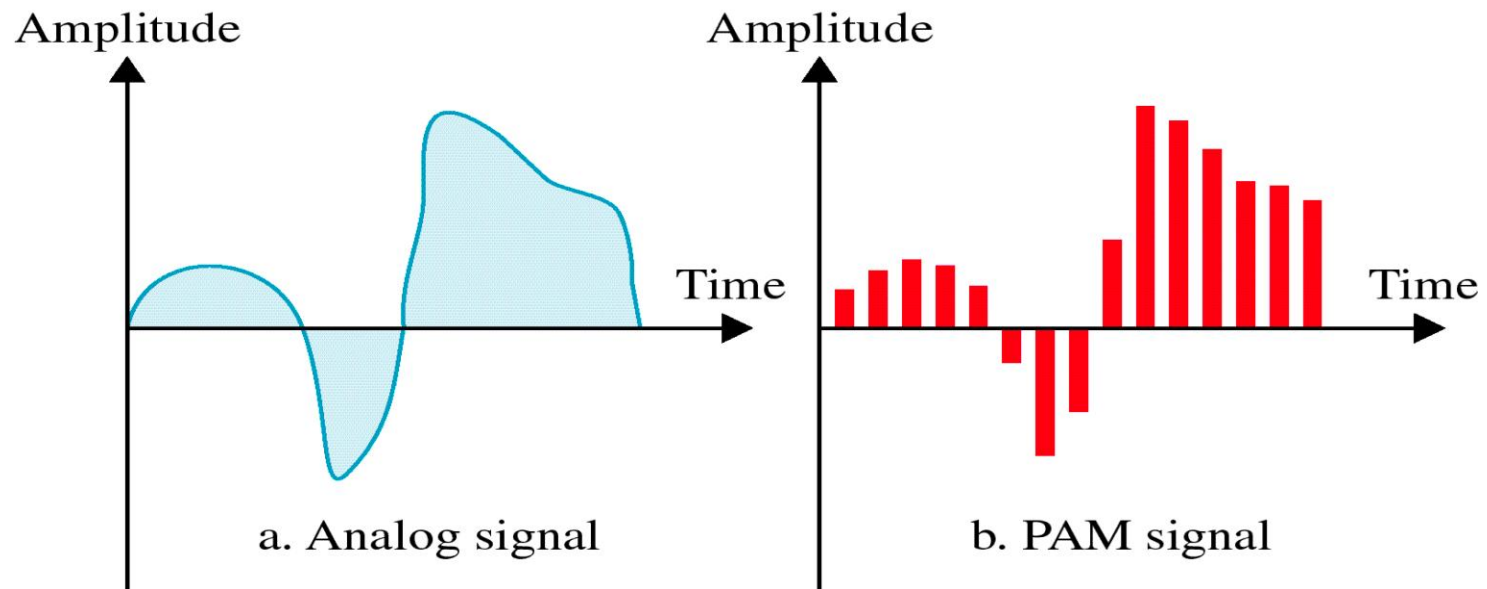


Analog to Digital Encoding



Pulse Amplitude Modulation (PAM)

- This technique takes an analog signal, sample it and generates a series of pulses on the result of sampling.
- **Sampling:** Measuring the amplitude of the signal at equal interval.



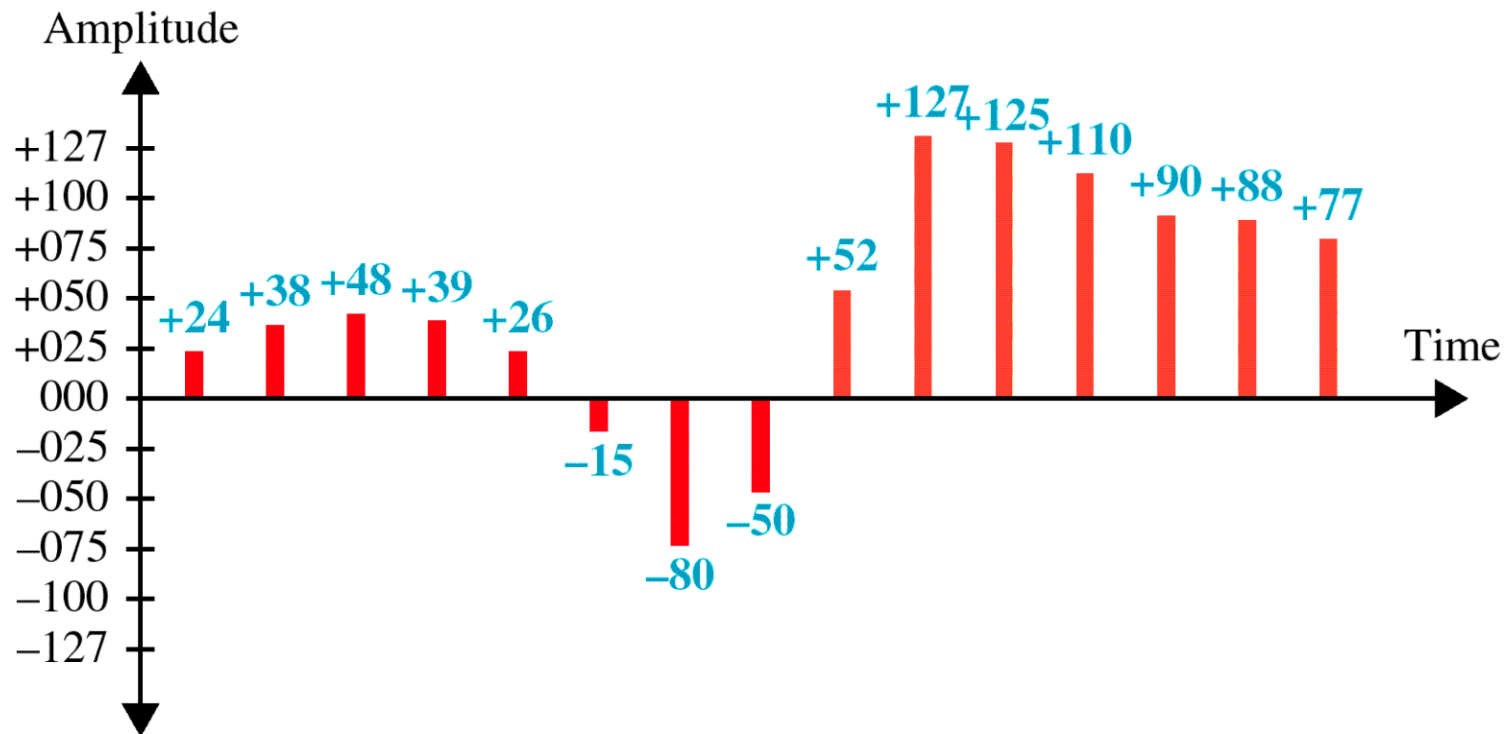
PAM has some applications, but is not used by itself in data communication. However, it is the first step in another very popular conversion method called **pulse code modulation (PCM)**

PCM Pulse Code Modulation

PCM is made up of four separate processes: PAM, quantization, binary encoding and digital to digital encoding.

Quantized PAM Signal

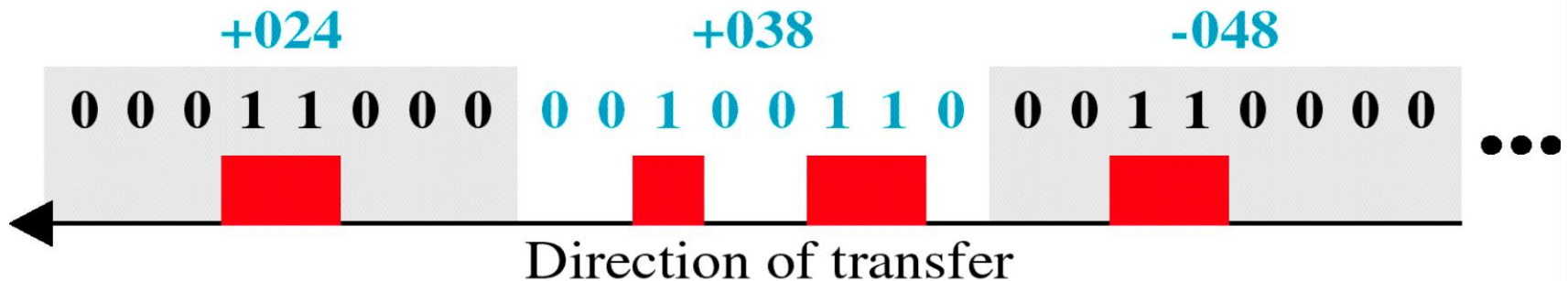
Quantization is a method of assigning the value of amplitude.



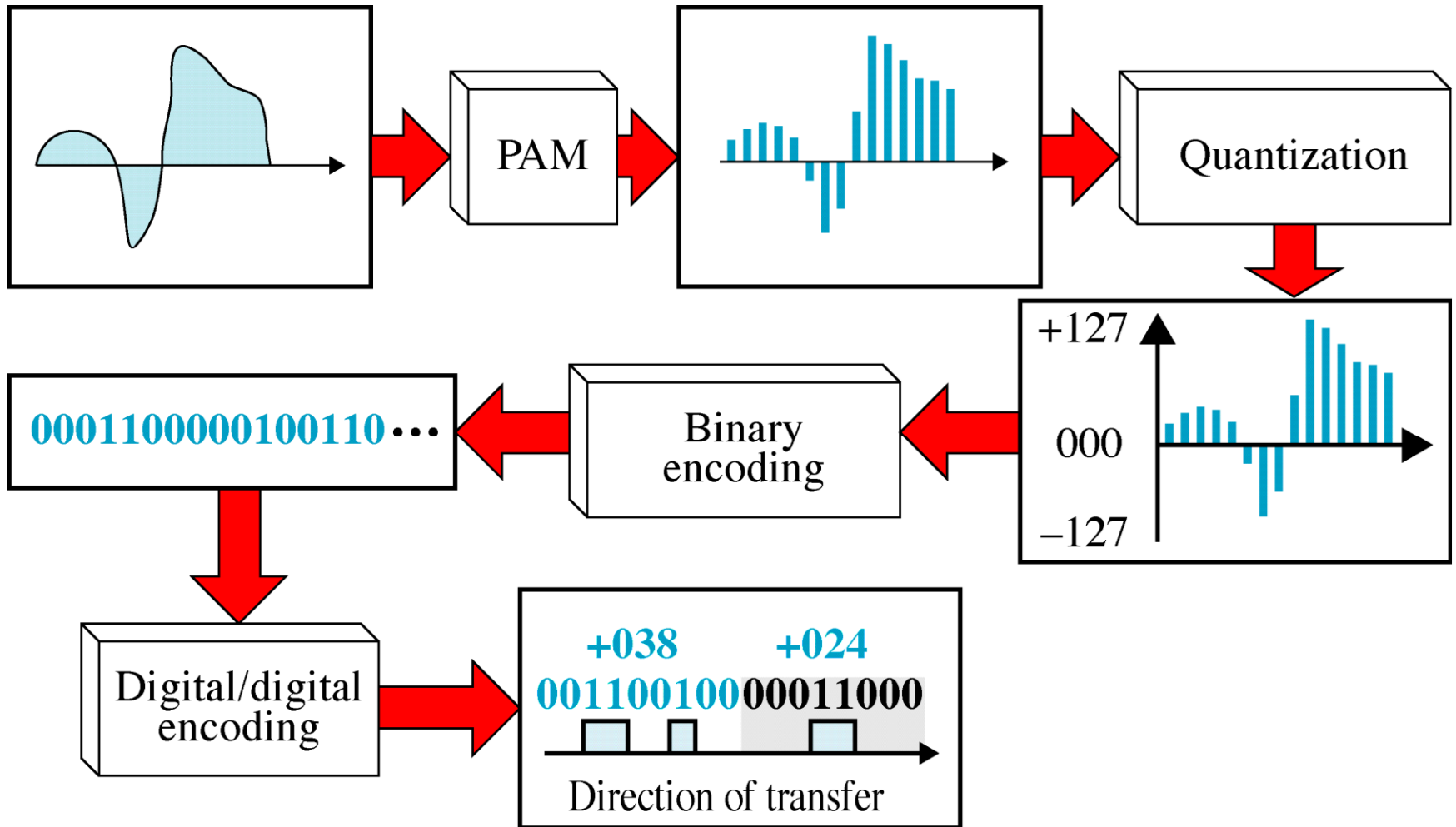
Binary Encoding: Quantizing Using Sign and Magnitude

+024	00011000	-015	10001111	+125	01111101
+038	00100110	-080	11010000	+110	01101110
+048	00110000	-050	10110010	+090	01011010
+039	00100111	+052	00110110	+088	01011000
+026	00011010	+127	01111111	+077	01001101

Sign bit
+ is 0 - is 1

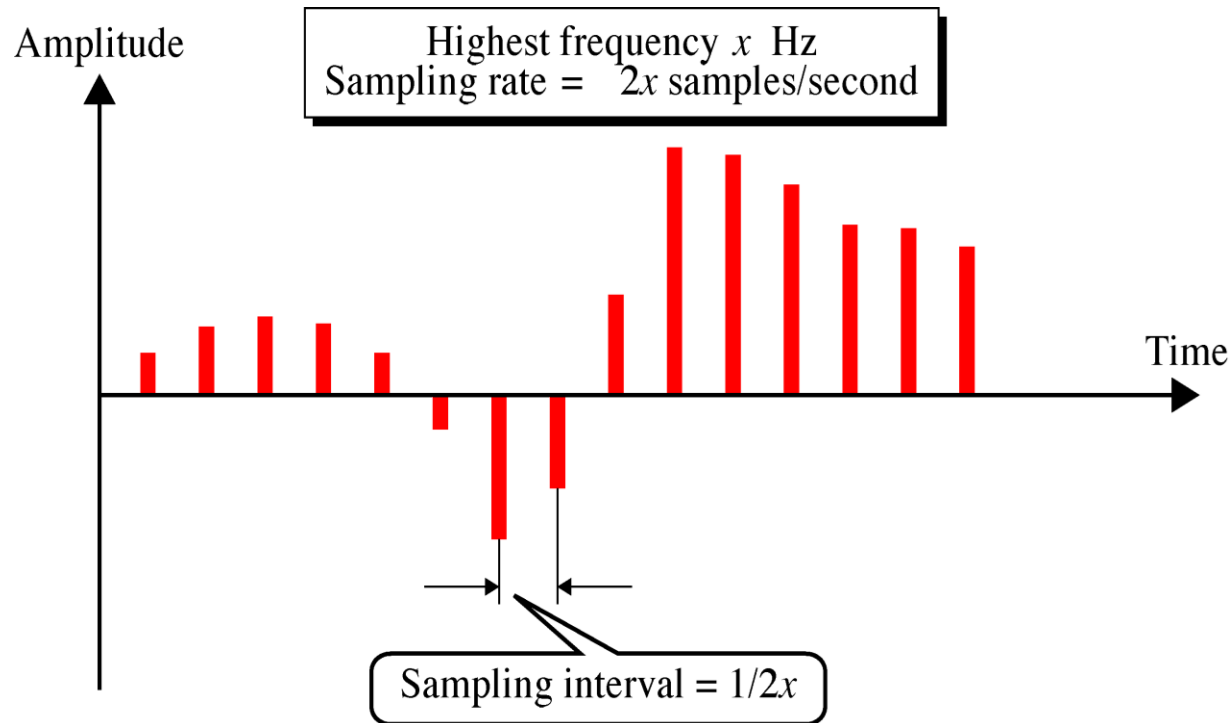


From Analog to PCM



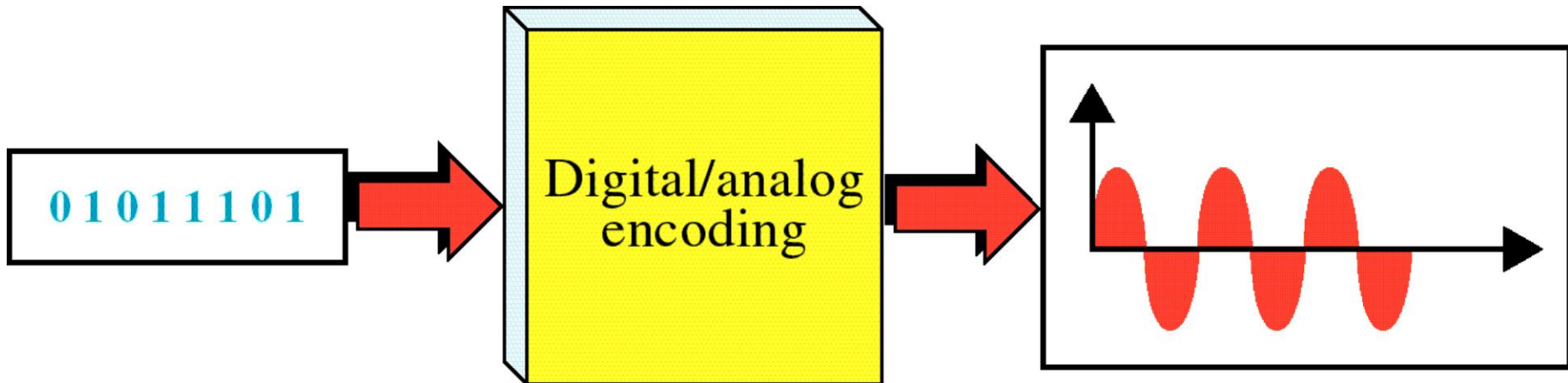
Nyquist Theorem

To ensure the accurate reproduction of an original analog signal using PAM, the **sampling rate** must be at least two times the highest frequency.



$$\text{Bit rate} = \text{Sampling rate} * \text{Number of bits per sample}$$

Digital to Analog Encoding



Bit Rate: Number of bits transmitted during one second.

Baud Rate: refers to the number of signal units per second that are required to represent those bits.

Baud rate is less than or equal to the bit rate.

Bit rate = baud rate * no of bits per signal element.

Carrier Signal (Base Signal): The sending device produces a high-frequency signal that acts as a basis for the information signal. This base signal is called the carrier signal.

Shift Keying: (Modulation): Digital information is modulated on the carrier signal by modifying one or more of its parameters (amplitude, frequency, phase). This kind of modification is called shift keying.

There are three types of shift keying: **amplitude shift keying (ASK)**, **frequency shift keying (FSK)** and **phase shift keying (PSK)**.

There is a fourth type of modulation: **quadrature amplitude modulation (QAM)**

Digital/analog
encoding

```
graph TD; A[Digital/analog encoding] --> B[ASK]; A --> C[FSK]; A --> D[PSK]; B -.-> E[QAM]; D -.-> E;
```

ASK

FSK

PSK

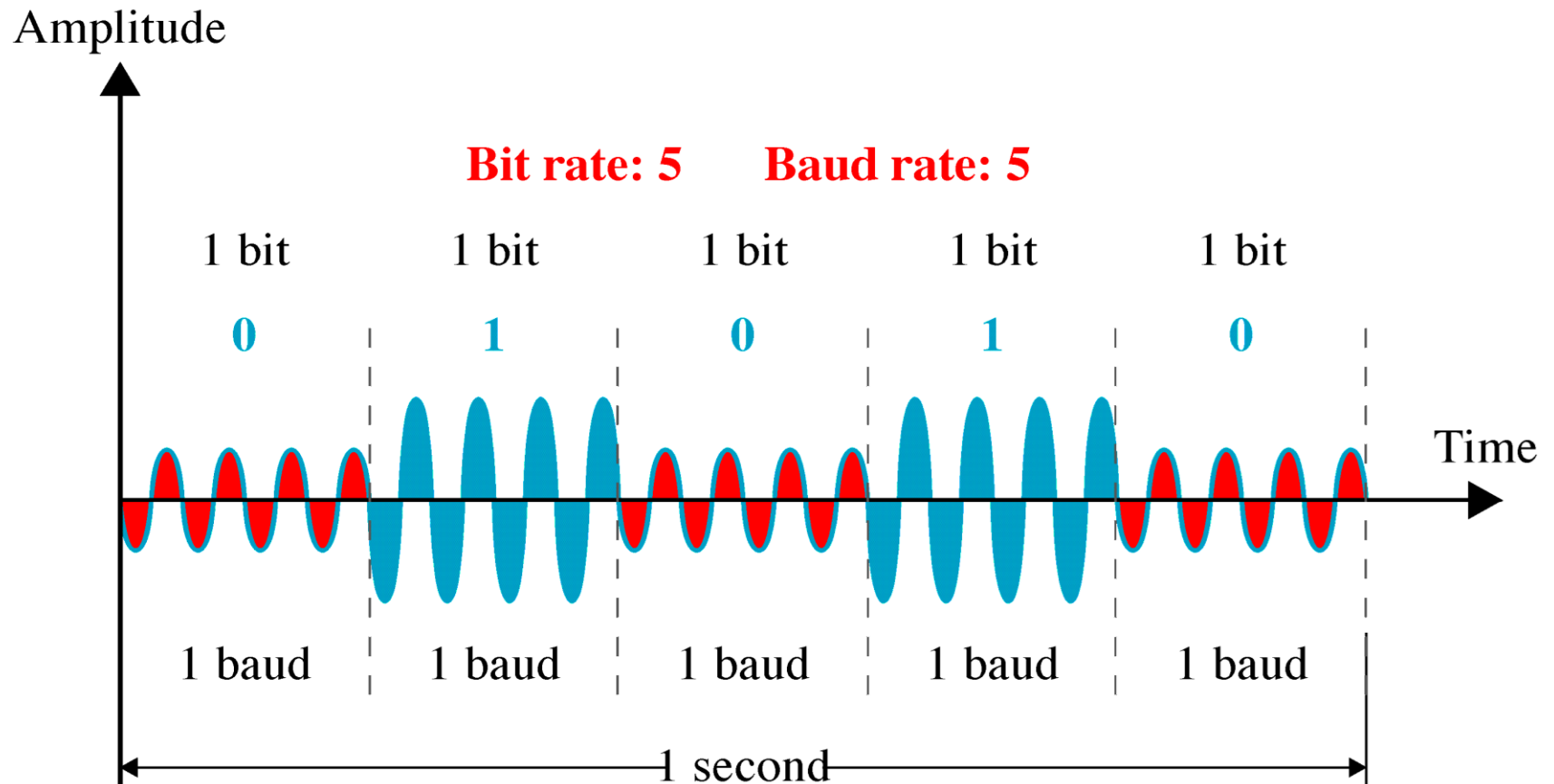
QAM

ASK

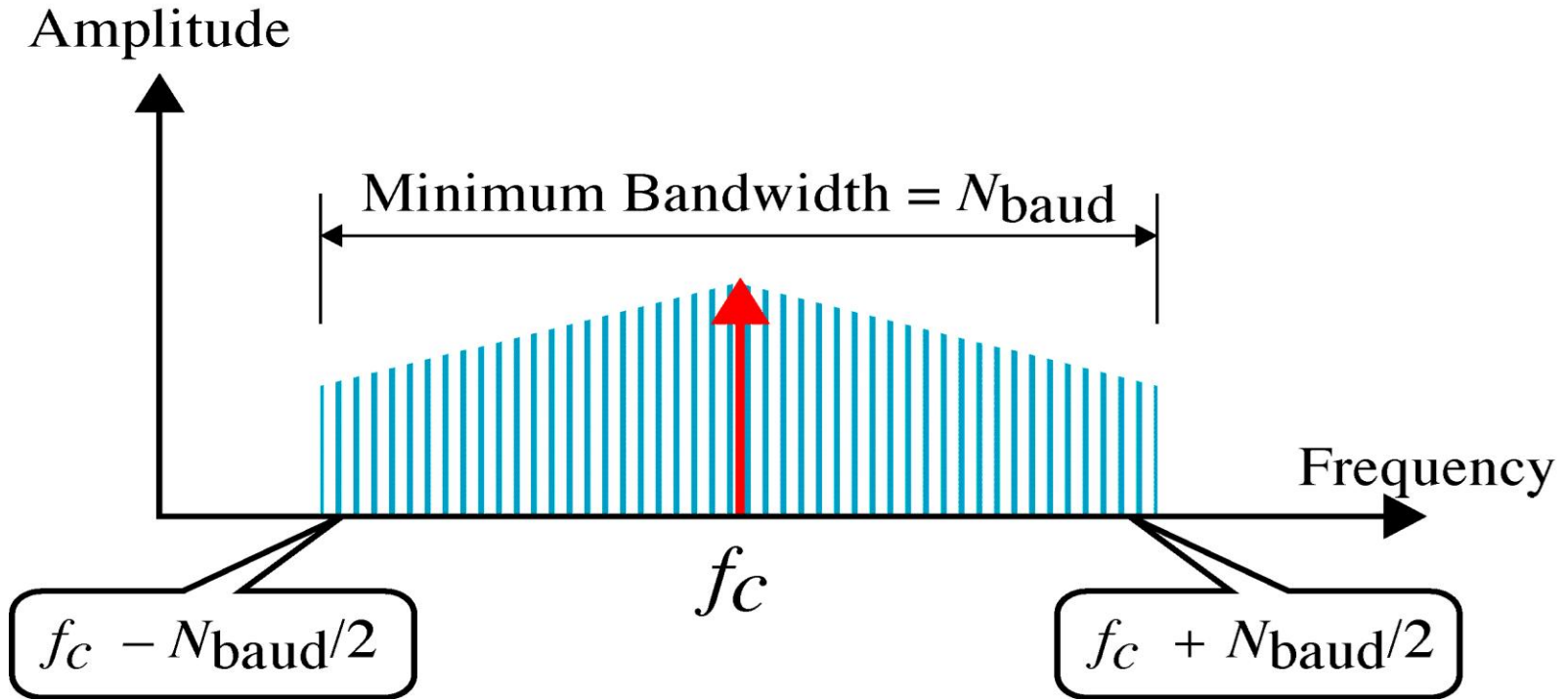
Amplitude varied to represent binary 1 or 0. Both frequency and phase remain constant while the amplitude changes.

The peak amplitude of the signal during each bit duration is constant and its value depends on the bit (0 or 1).

ASK is the modulating method most affected by noise.



Bandwidth for ASK

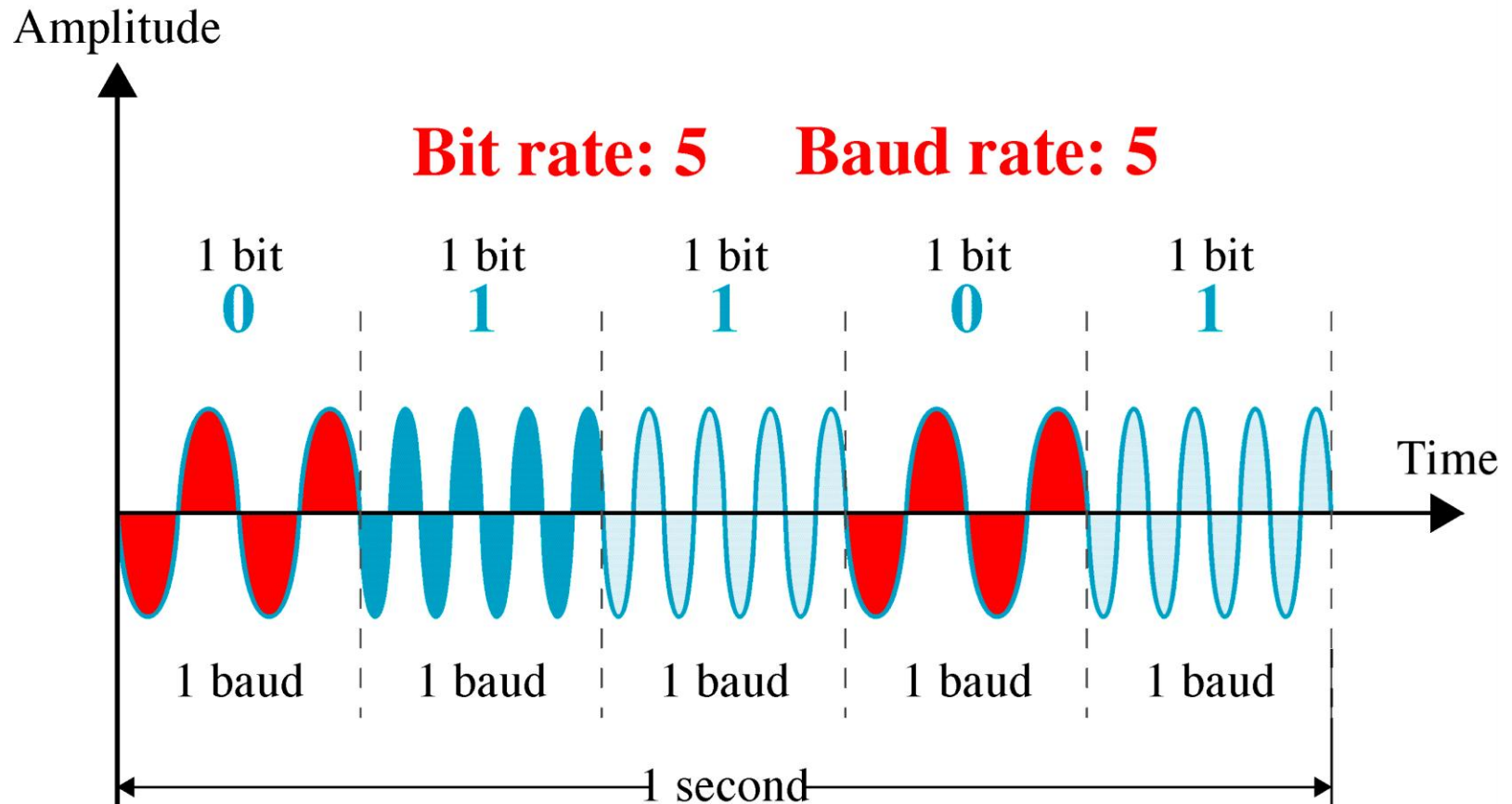


$$\text{Bandwidth for ASK} = (1 + d) * N_{\text{baud}}$$

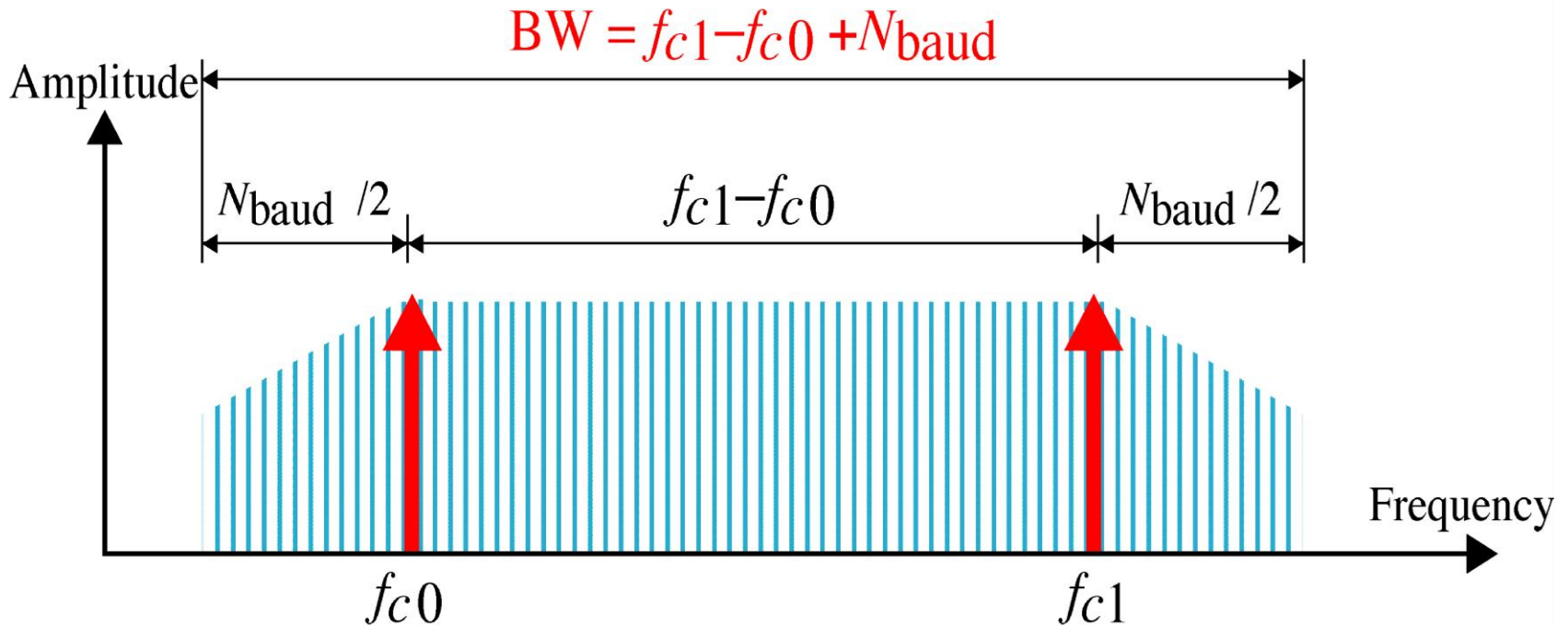
d: factor related to the condition of the line (with a minimum value 0)

FSK

The frequency of the carrier signal is varied to represent binary 1 or 0. Both peak amplitude and phase remain constant. The frequency of the signal during each bit duration is constant and its value depends on the bit (0 or 1).



Bandwidth for FSK

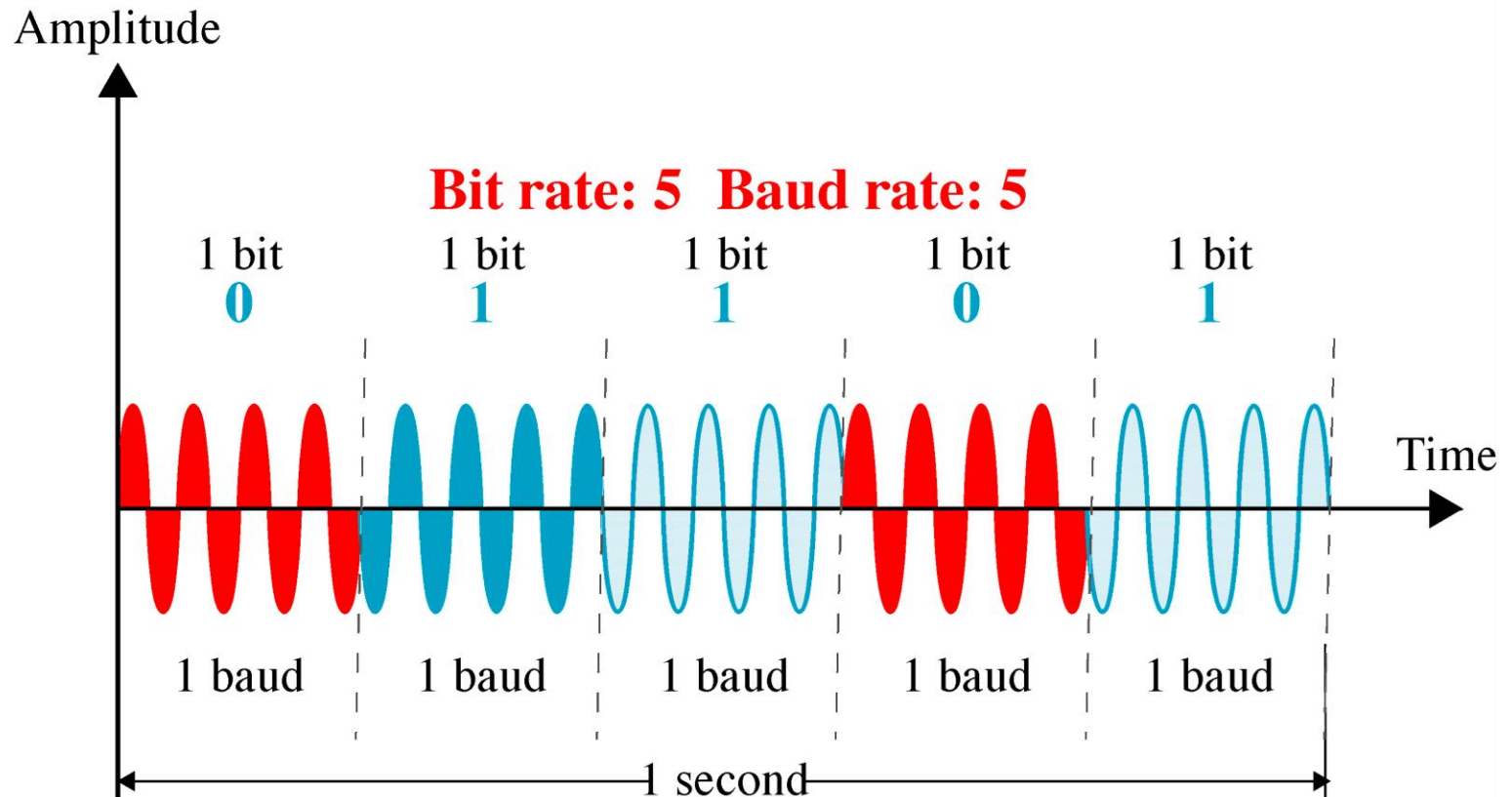


FSK avoids most of the noise problems of ASK. Because the receiving device is looking for specific frequency changes over a given number of periods. It can ignore voltage spike.

PSK

The phase of the carrier signal is varied to represent binary 1 or 0. Both peak amplitude and frequency remain constant. The phase of the signal during each bit duration is constant and its value depends on the bit (0 or 1).

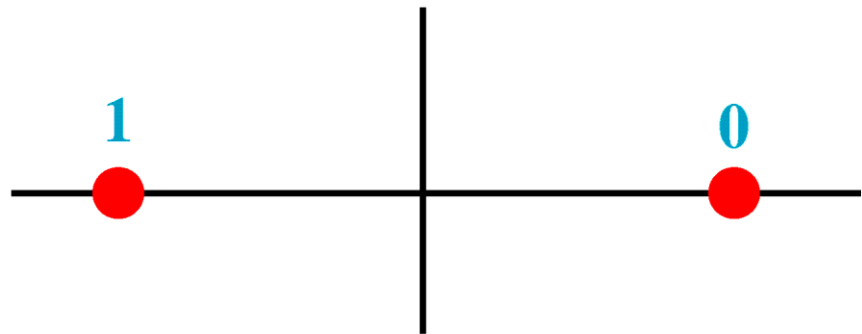
For example: a phase of 0 degree represent binary 0 and phase of 180 degree represent binary 1. it uses two phases called 2-PSK.



PSK Constellation

Bit	Phase
0	0
1	180

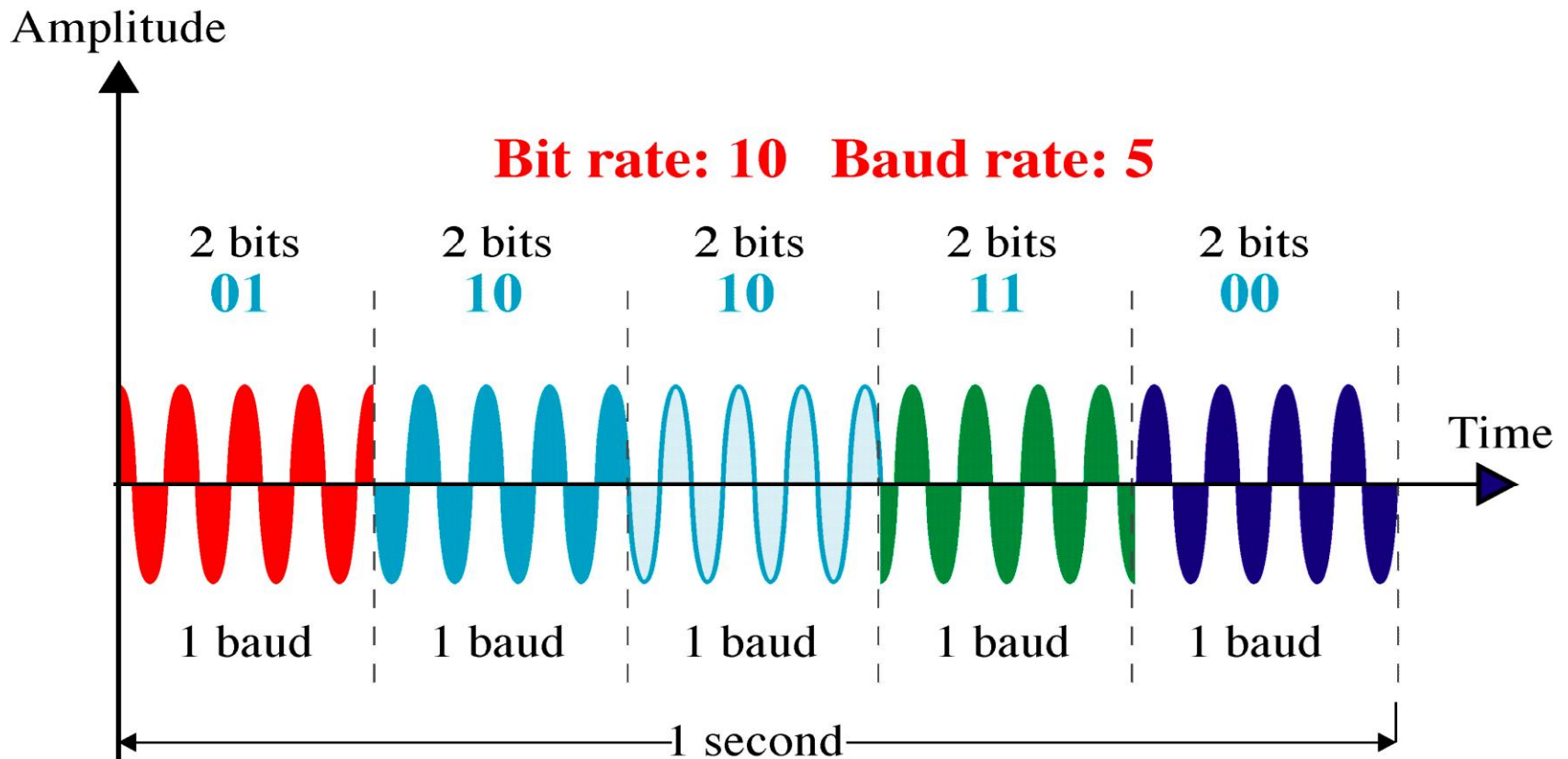
Bits



Constellation diagram

4-PSK

This example uses four phases: a phase of 0 degree represent binary 00, phase of 90 degree represent binary 01, phase of 180 degree represent binary 10 and phase of 270 degree represent binary 11. it uses four phases called 4-PSK.

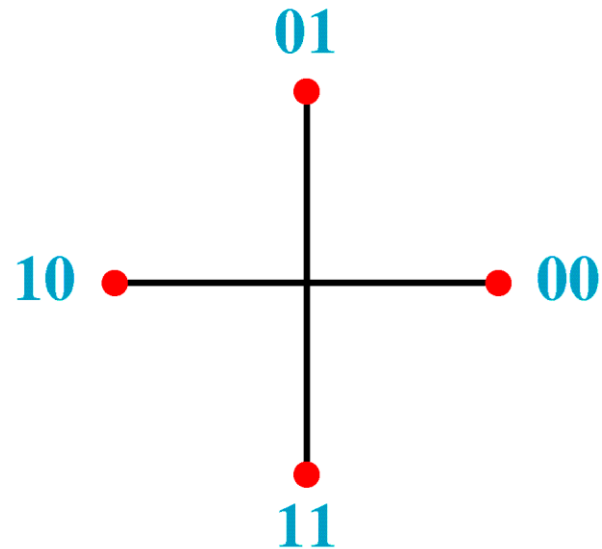


4-PSK

Characteristics

Dibit	Phase
00	0
01	90
10	180
11	270

Dibit
(2 bits)



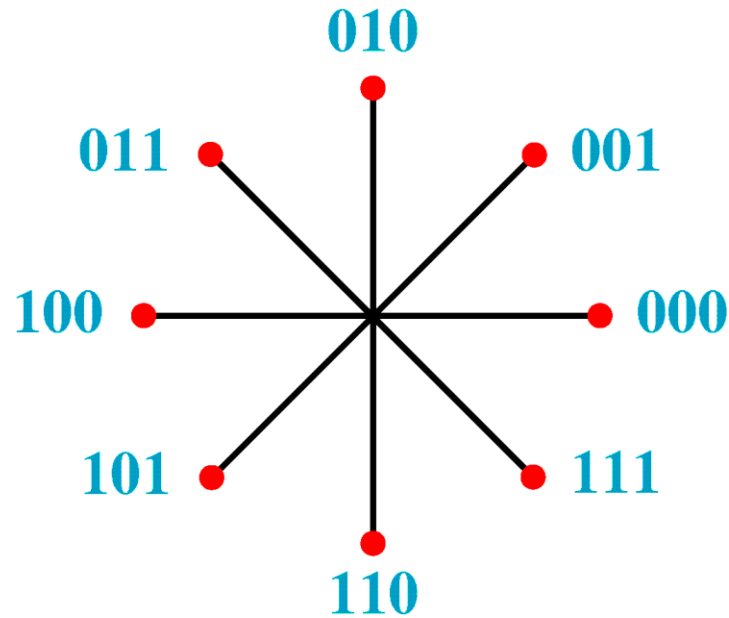
Constellation diagram

8-PSK

Characteristics

Tribit	Phase
000	0
001	45
010	90
011	135
100	180
101	225
110	270
111	315

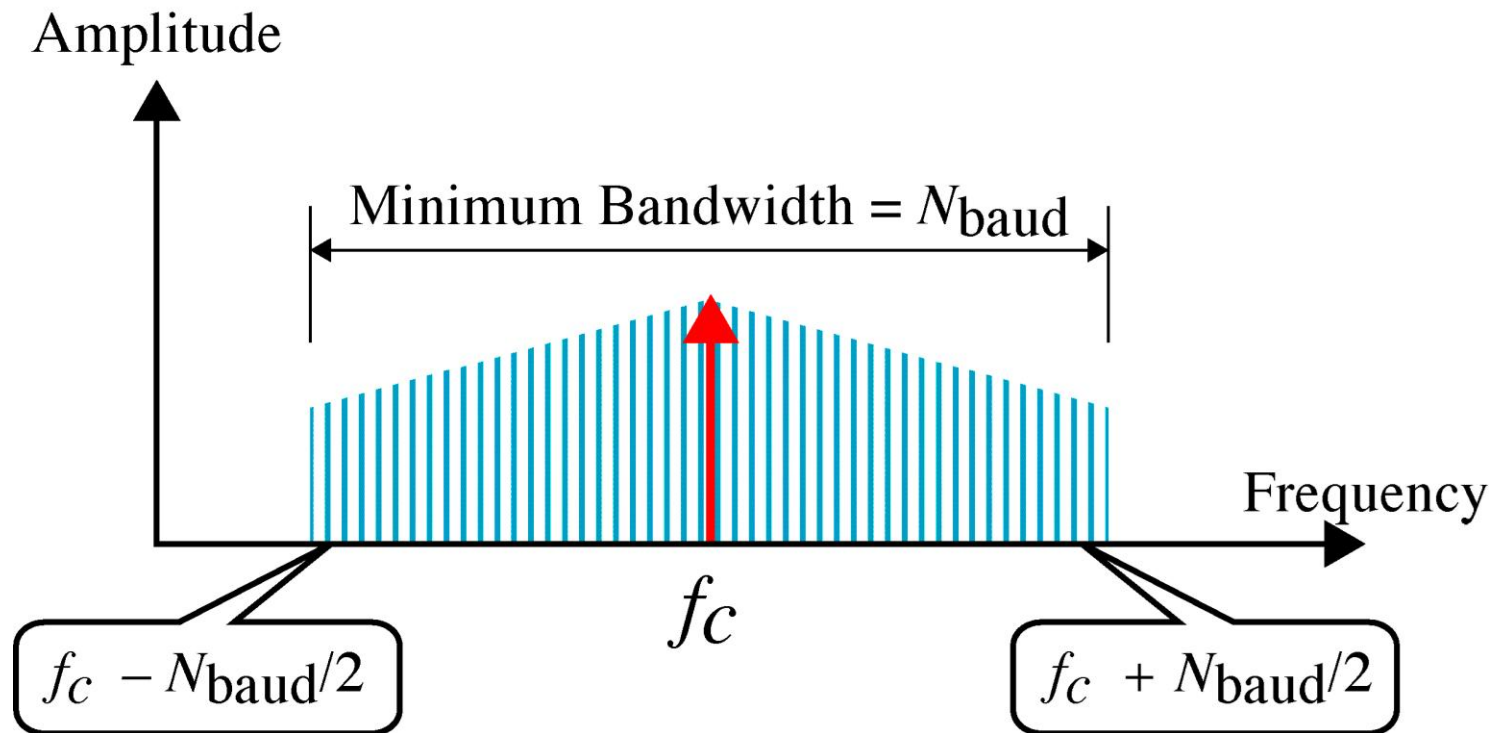
Tribits
(3 bits)



Constellation diagram

PSK

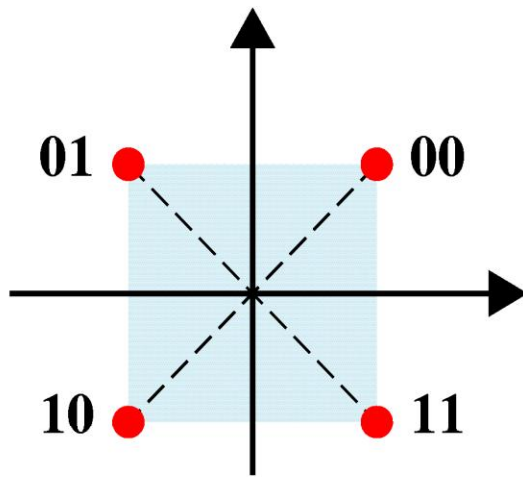
Bandwidth



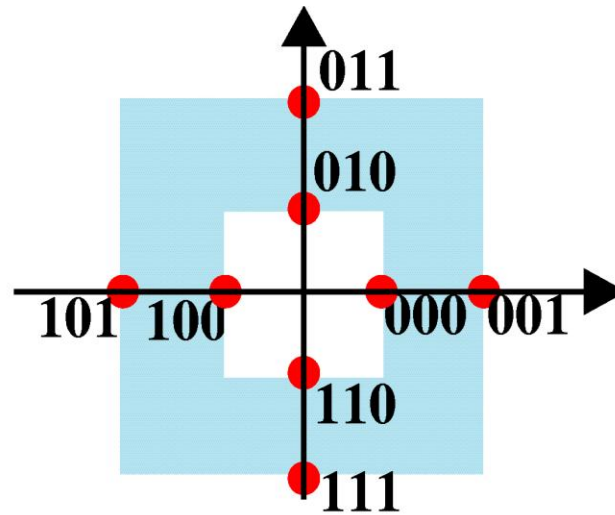
Quadrature Amplitude Modulation (QAM)

QAM is a combination of ASK and PSK. Theoretically, any measurable number of changes in amplitude can be combined with any number of changes in phase.

4-QAM and 8-QAM Constellations

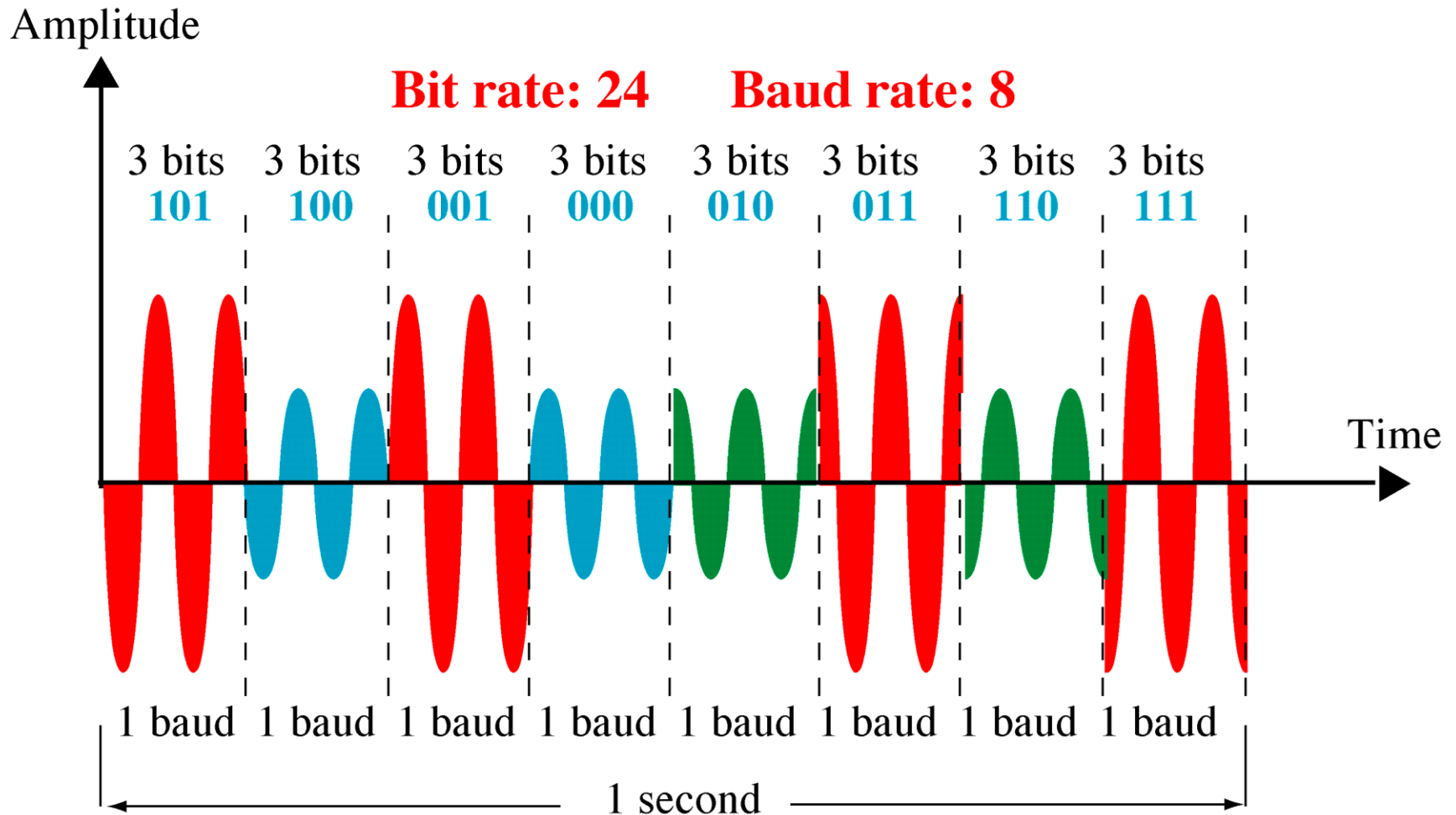


4-QAM
1 amplitude, 4 phases



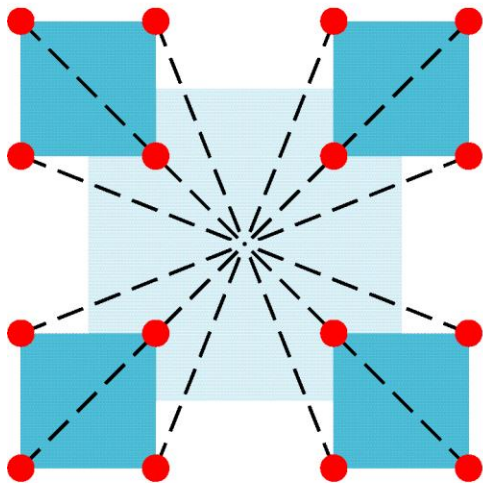
8-QAM
2 amplitudes, 4 phases

8-QAM Signal



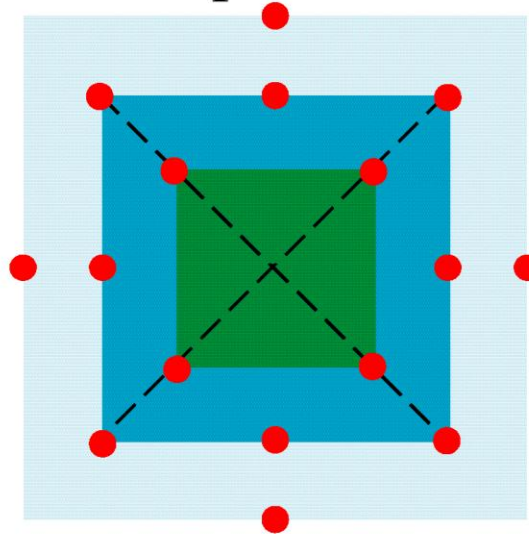
16-QAM Constellation

3 amplitudes,
12 phases



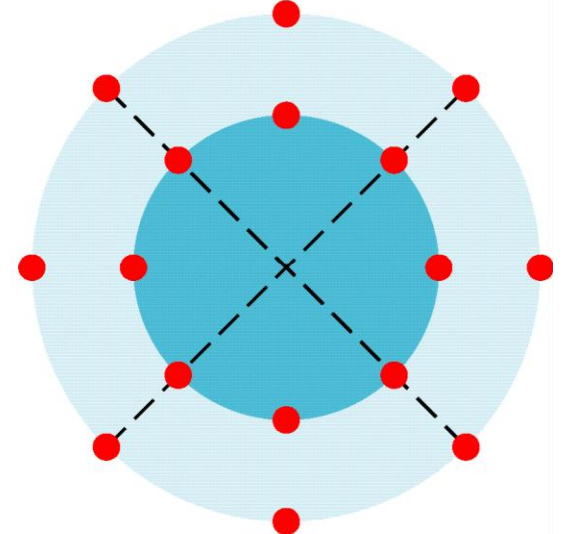
16-QAM

4 amplitudes,
8 phases



16-QAM

2 amplitudes,
8 phases



16-QAM

Bit Rate and Baud Rate

Bit

Baud rate = N

Bit rate = N

0	0	1	0	1	0	0	0	1	0	1	0	1	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Dibit

Baud rate = N

Bit rate = $2N$

0	0	1	0	1	0	0	0	1	0	1	0	1	1	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Tribit

Baud rate = N

Bit rate = $3N$

0	0	1	0	1	0	0	0	1	0	1	0	1	1	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

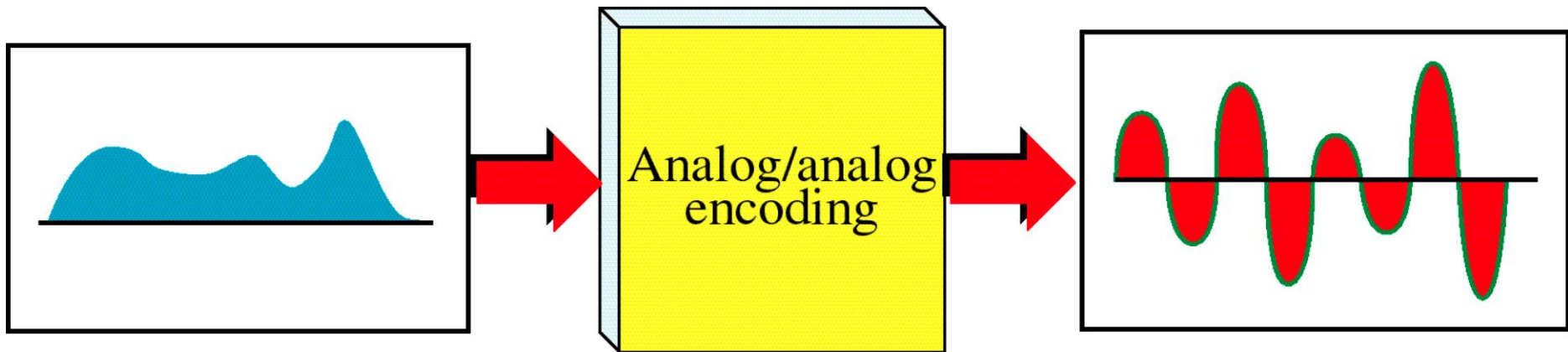
Quadbit

Baud rate = N

Bit rate = $4N$

0	0	1	0	1	0	0	0	1	0	1	0	1	1	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Analog to Analog Modulation



Analog/analog
encoding

```
graph TD; A[Analog/analog encoding] --> B[AM]; A --> C[FM]; A --> D[PM];
```

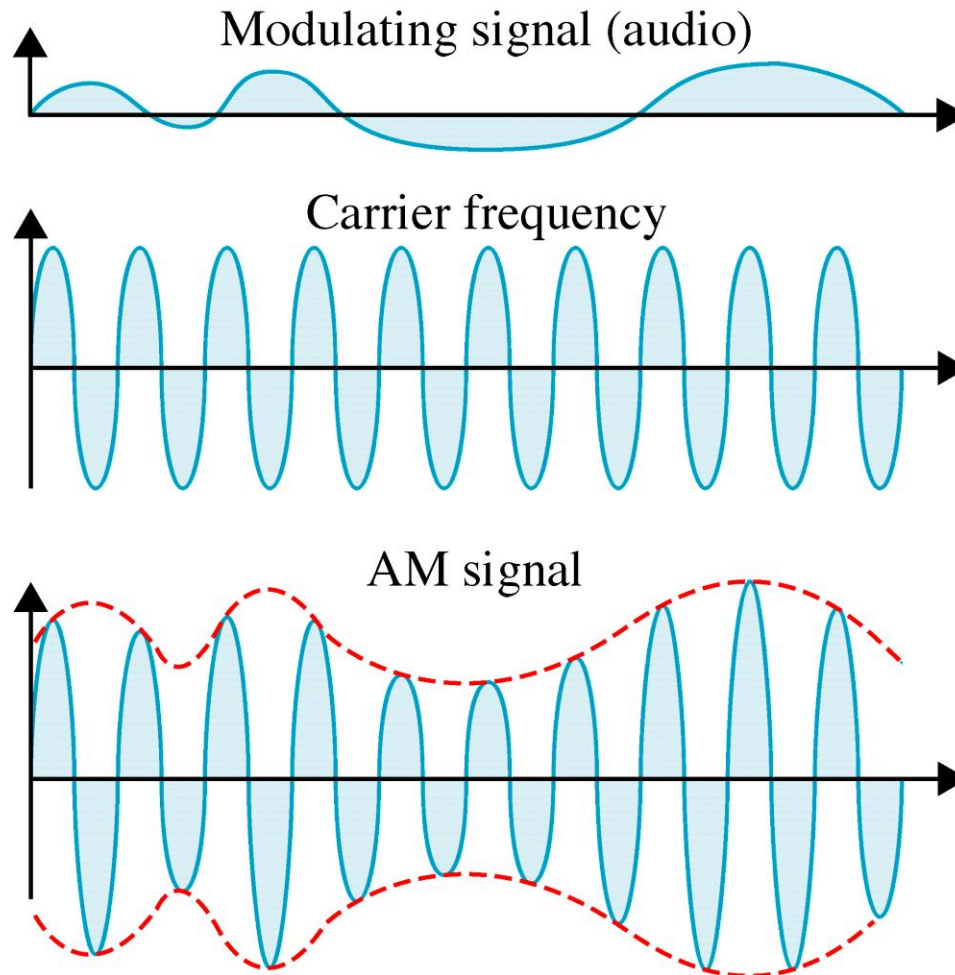
AM

FM

PM

Amplitude Modulation

The carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal. The frequency and phase of the carrier remain the same; only the amplitude changes to follow variations in the information.

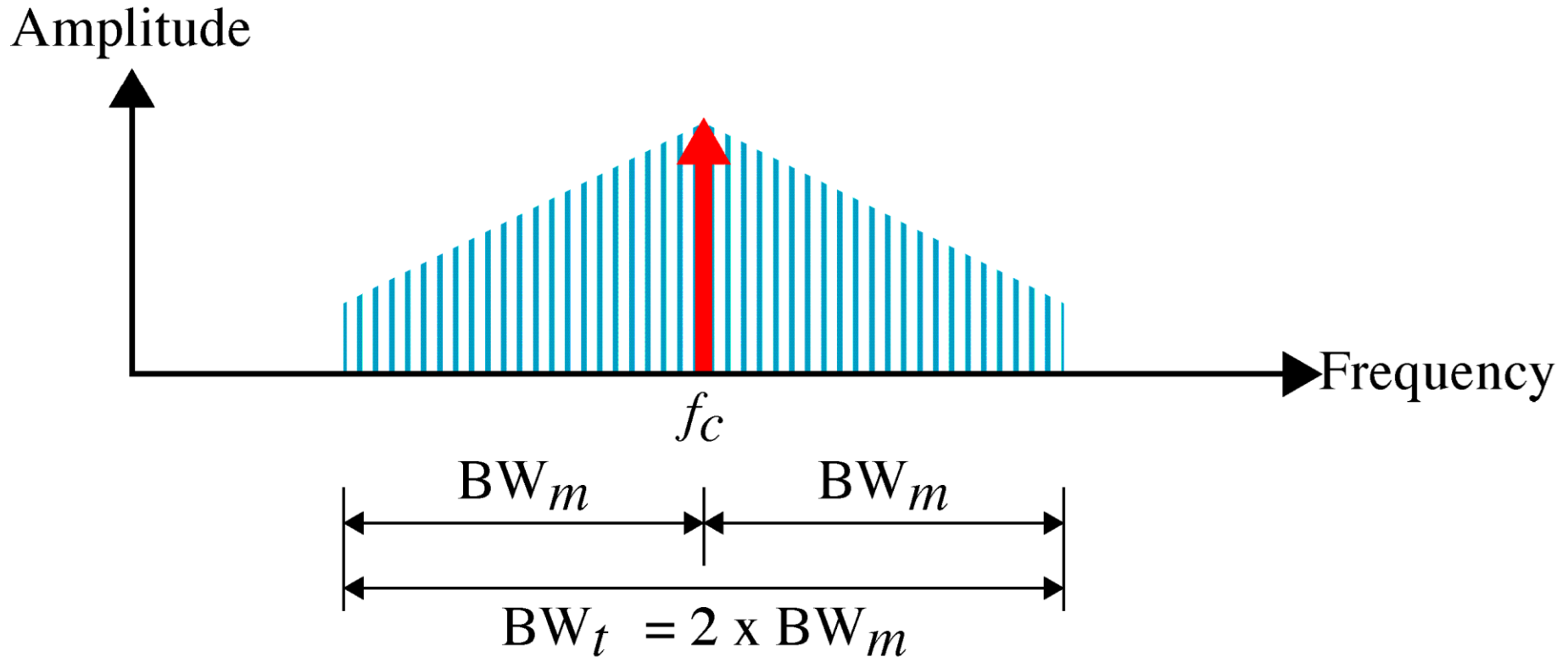


AM Bandwidth

BW_m = Bandwidth of the modulating signal (audio)

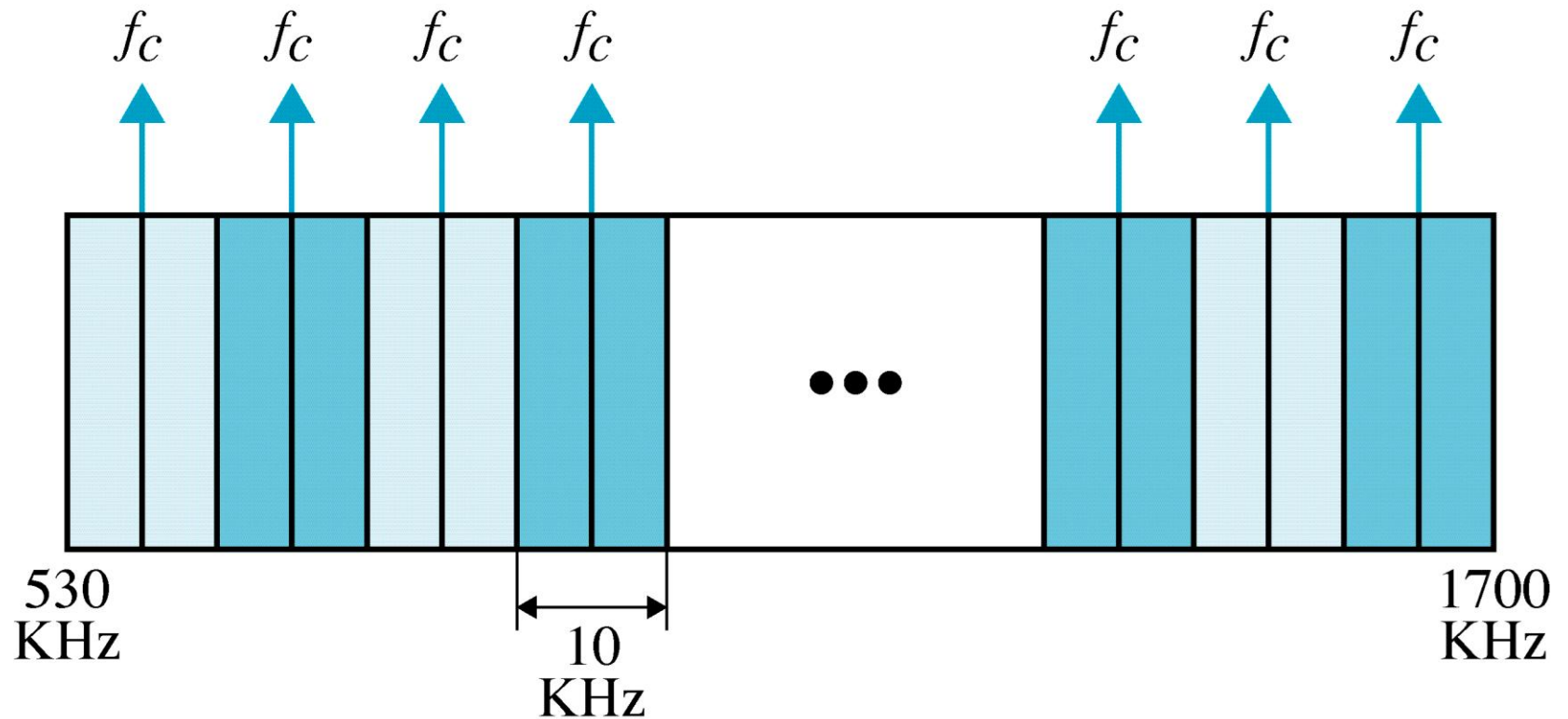
BW_t = Total bandwidth (radio)

f_c = Frequency of the carrier



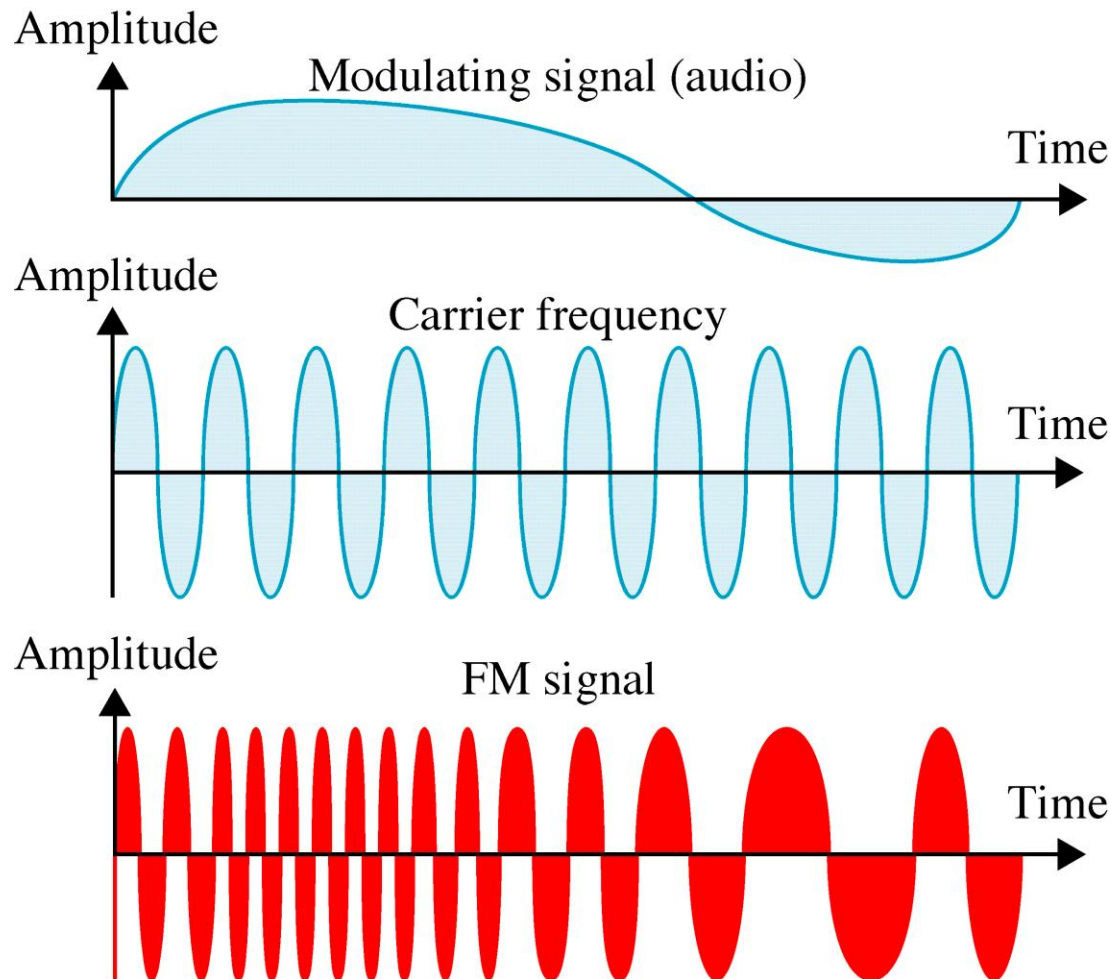
AM Band Allocation

f_c = Carrier frequency of the station



Frequency Modulation

The frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and phase of the carrier remain constant, but as the amplitude of the information signal changes, the frequency of the carrier changes correspondingly.



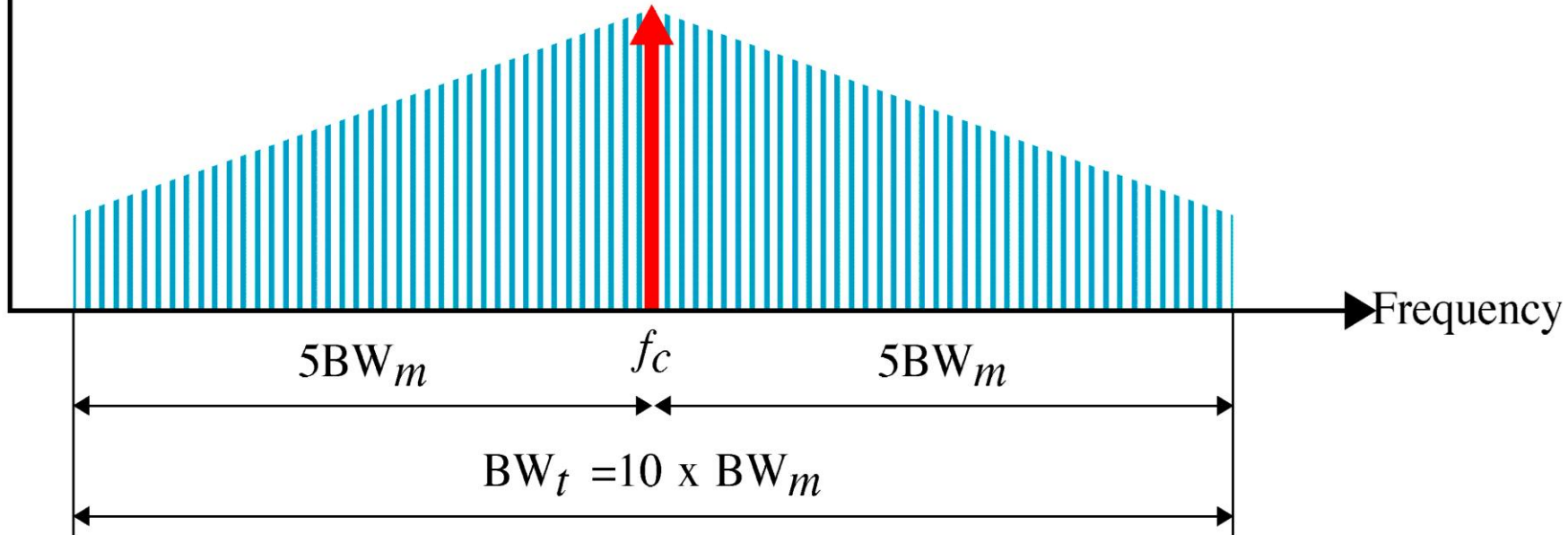
FM Bandwidth

Amplitude

BW_m = Bandwidth of the modulating signal (audio)

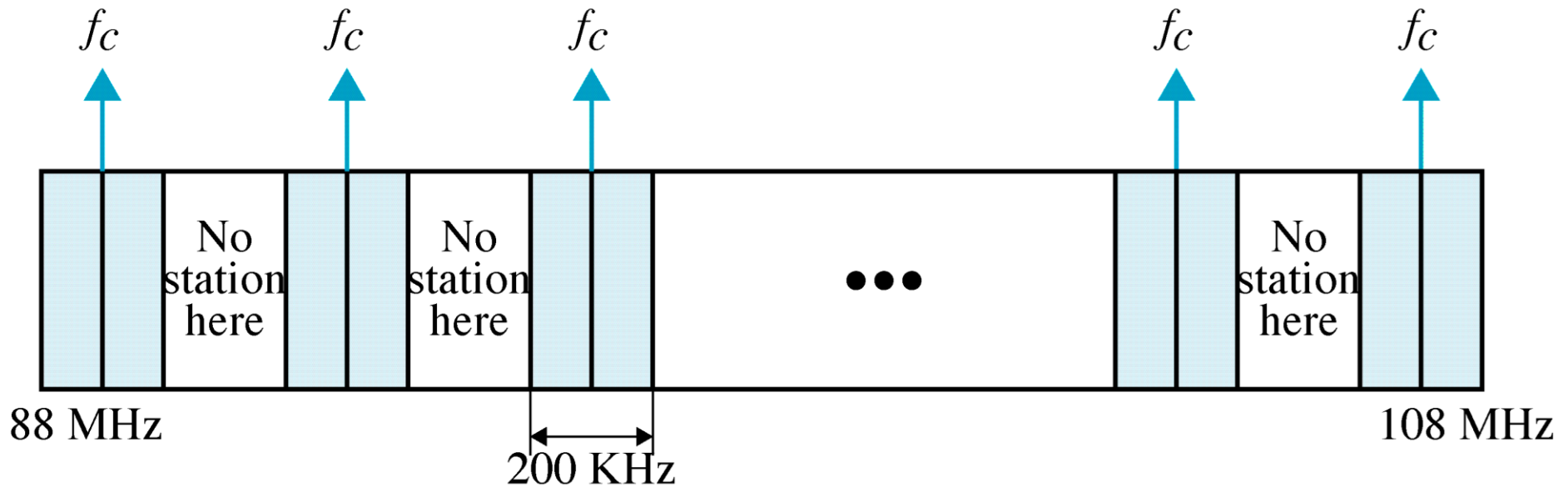
BW_t = Total bandwidth (radio)

f_c = Frequency of the carrier



FM Band Allocation

f_c = Carrier frequency of the station



Phase Modulation

The phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and frequency of the carrier remain constant, but as the amplitude of the information signal changes, the phase of the carrier changes correspondingly.